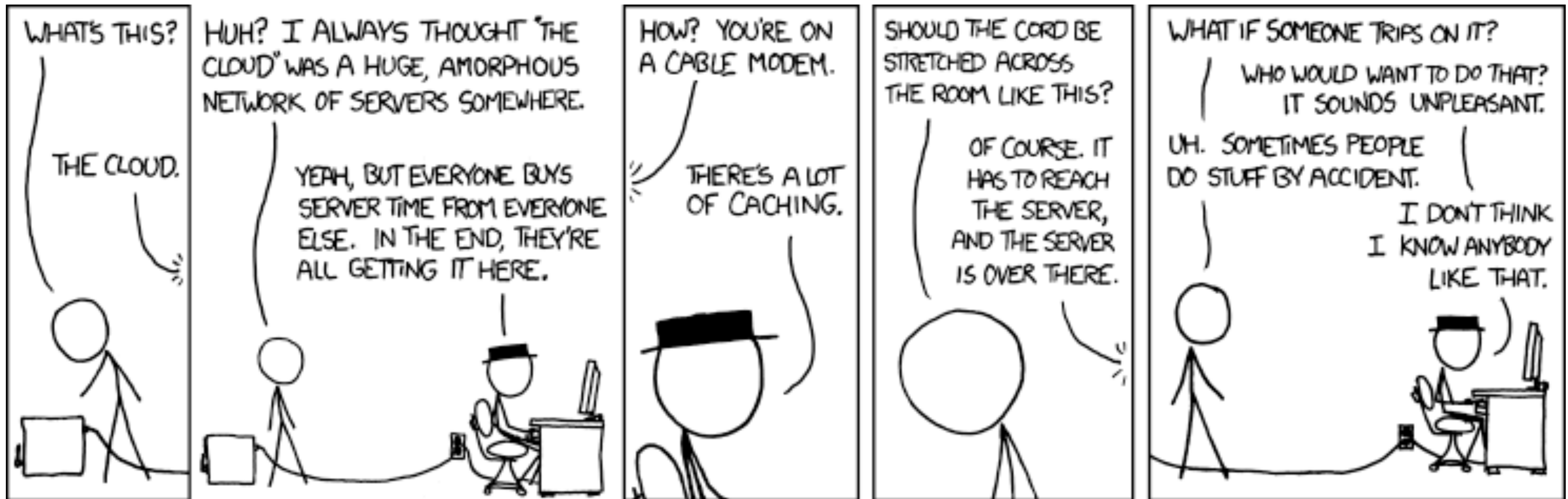


Networking Fundamentals



Randall Munroe
<https://xkcd.com/908/>

Learning Goals

- ▶ Get an understanding of what a computer network is
 - ▶ Network Topology
 - ▶ The network edge
 - ▶ Access networks
 - ▶ The network core
 - ▶ Forwarding & Routing
- ▶ Understand what a protocol is
 - ▶ Definition & Example
- ▶ Understand how layers form the structure of the Internet (and other networks)
 - ▶ General concepts of layering
 - ▶ The 5-layer Internet model
 - ▶ Services, Layering, and Encapsulation

Learning Goals

- ▶ Know how to measure the performance of computer networks
 - ▶ Delay
 - ▶ Loss
 - ▶ Throughput
- ▶ Understand the implications of the Internet's history
 - ▶ A brief intro to the history of the Internet
 - ▶ Implications for security

Learning Goal D88317

What is a Computer Network?

Network Definition by NIST

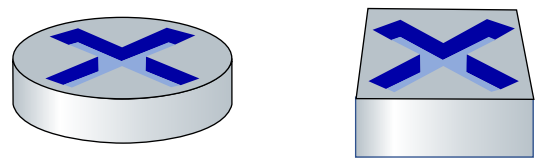
*A system implemented with a collection of interconnected components.
Such components may include routers, hubs, cabling,
telecommunications controllers, key distribution centres, and technical
control devices.*

- NIST SP 800-172

Network Topology



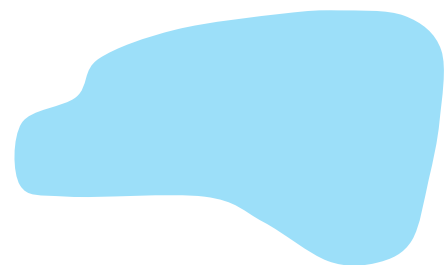
- ▶ Connected computing devices
 - ▶ Hosts = end systems
 - ▶ The network edge



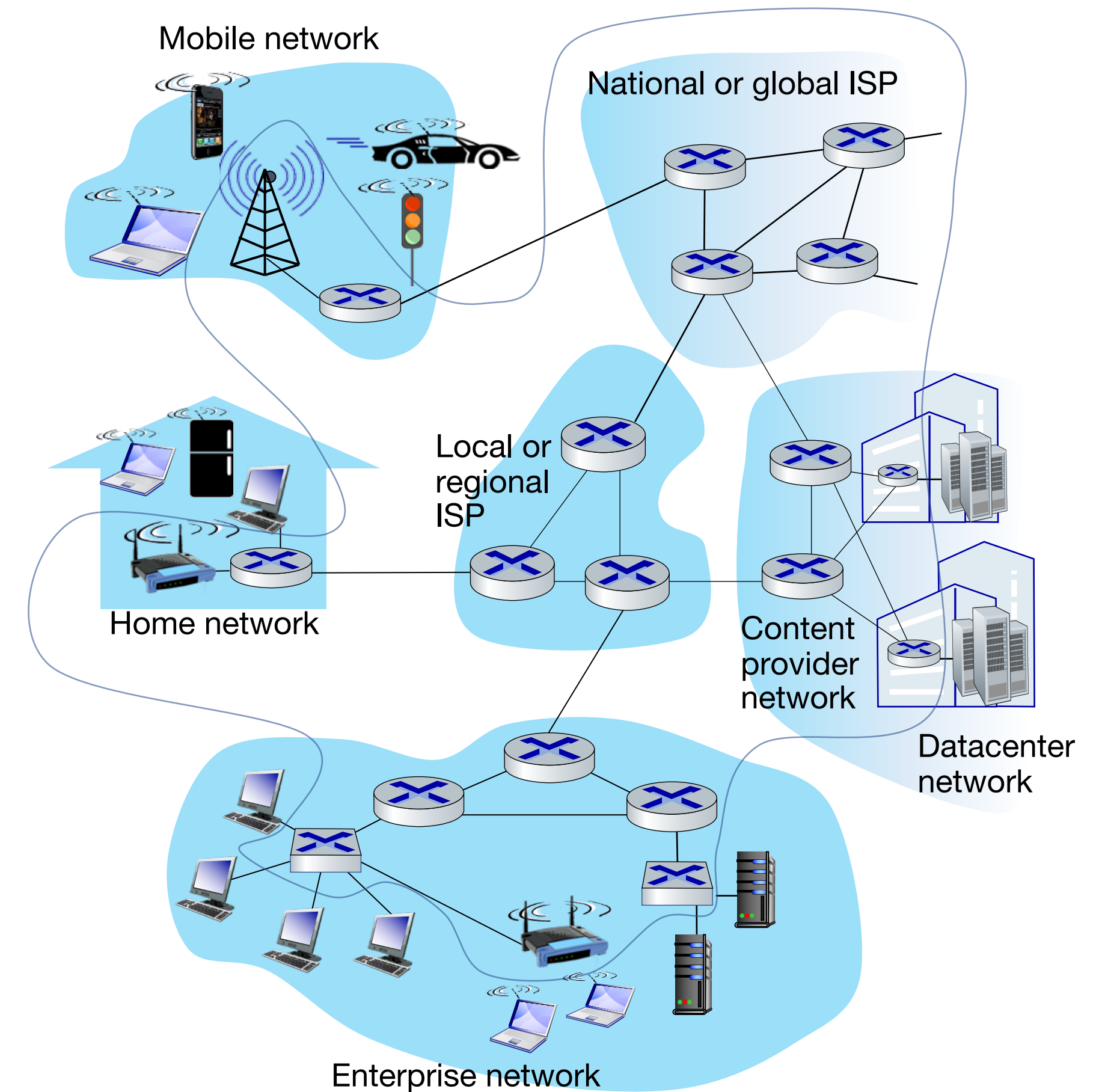
- ▶ Packet switches
 - ▶ Routers & switches forward data



- ▶ Communication links
 - ▶ Any medium: copper, radio, etc.
 - ▶ Transmission rate: bandwidth

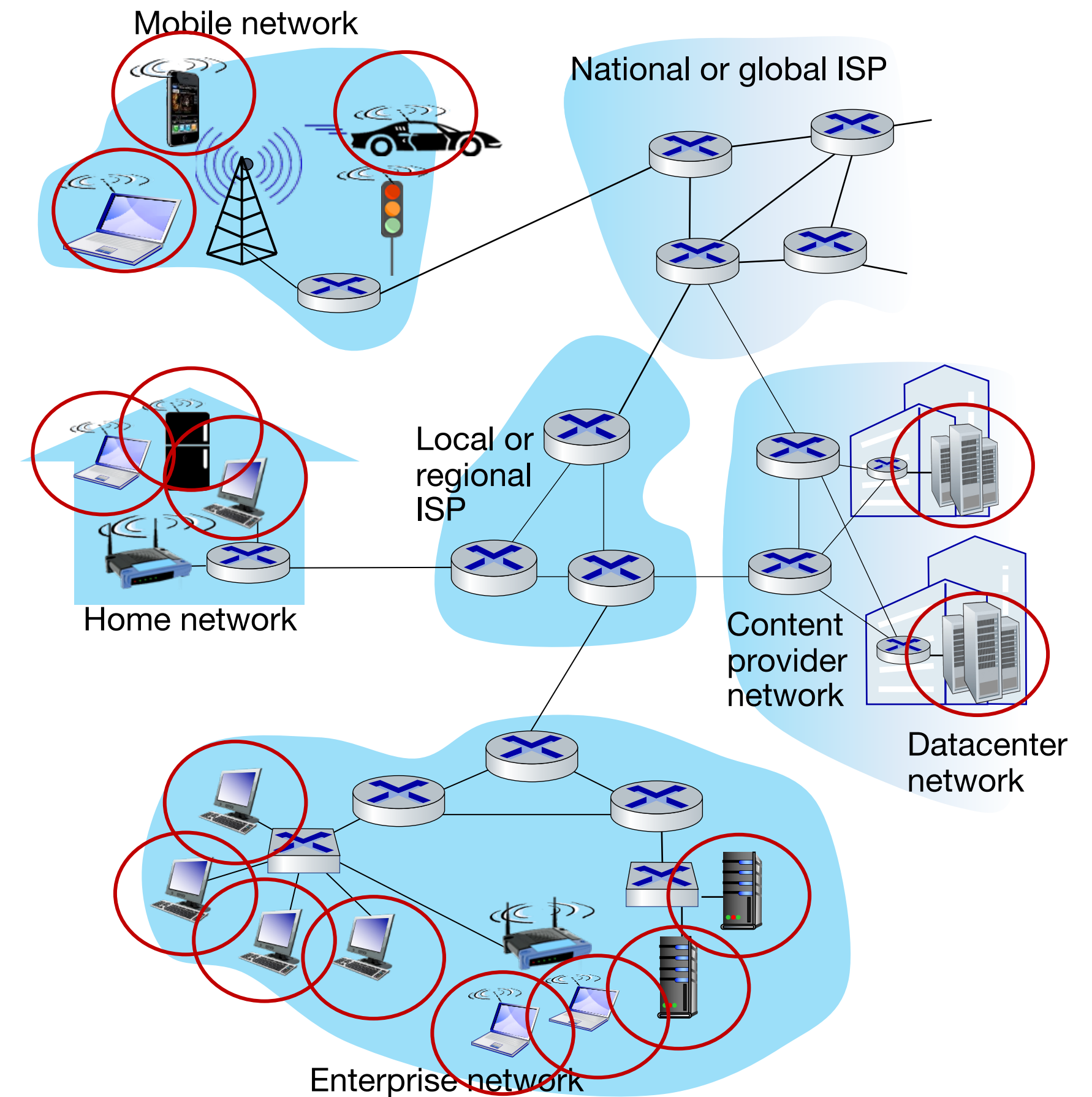


- ▶ Networks



Network Topology – The Network Edge

- ▶ The Network Edge
 - ▶ Hosts: clients and servers
 - ▶ Servers often in data centres



Hosts on the Network Edge

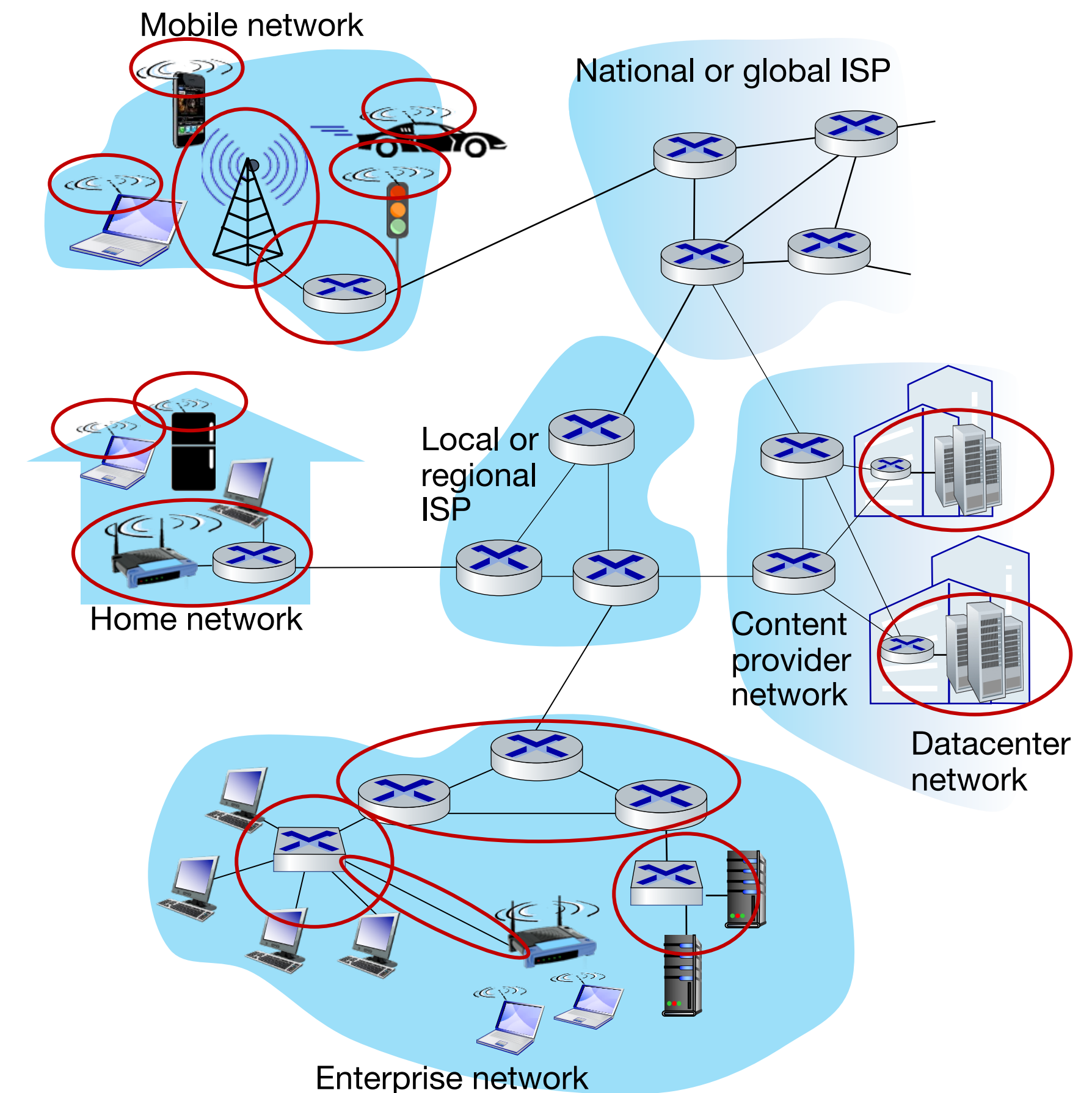
Traditional computing devices

Internet of things



Network Topology – Access Networks

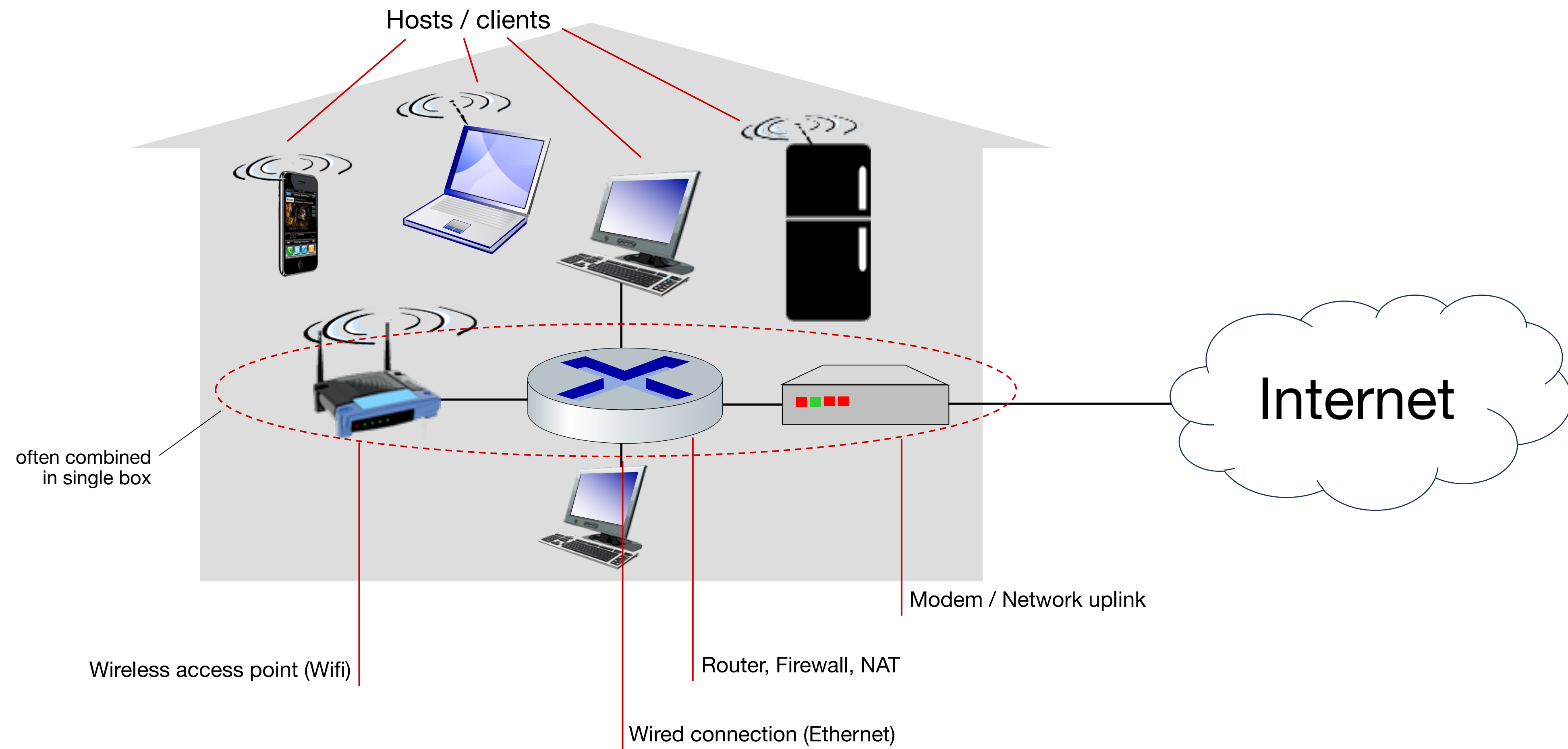
- ▶ The Network Edge
 - ▶ Hosts: clients and servers
 - ▶ Servers often in data centres
- ▶ Access Networks
 - ▶ Wired, wireless communication links
 - ▶ Provide connection to other networks through edge routers



A closer look at Access Networks

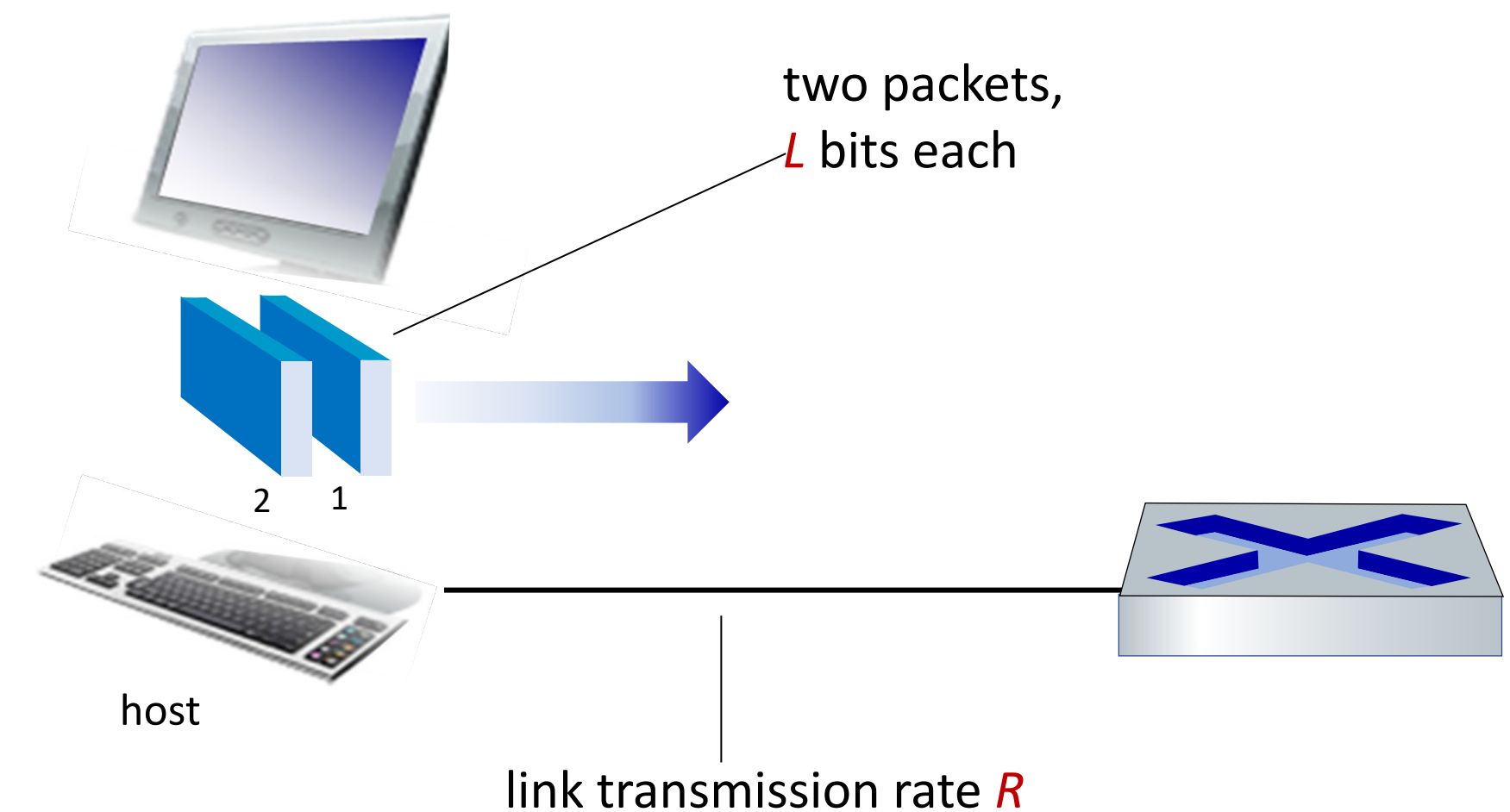
- ▶ How to connect hosts to the edge router?
 - ▶ Links
 - ▶ Wired or wireless access networks
 - ▶ Specifics for enterprise and data centre networks
- ▶ How to connect the edge router to the uplink?
 - ▶ Shared vs dedicated access

Interlude: The topology of a typical home network



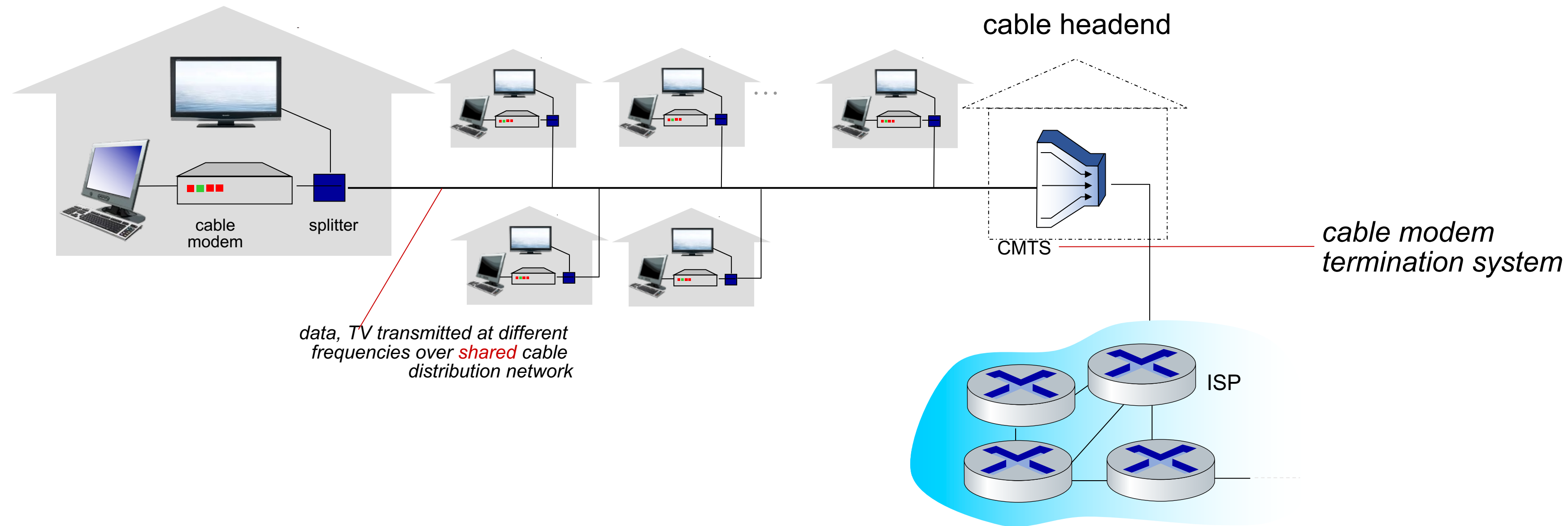
Links & Packets

- ▶ Packets are the logical structure of network communication
- ▶ Host takes the message
 - ▶ Breaks it into smaller chunks
 - ▶ These chunks are known as packets
 - ▶ Packets have a length in bits (L)
 - ▶ Packets are transmitted at rate R
 - ▶ Transmission rate also called capacity or link bandwidth



$$\text{packet transmission delay} = \frac{\text{time needed to transmit } L\text{-bit packet into link}}{L \text{ (bits)}} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

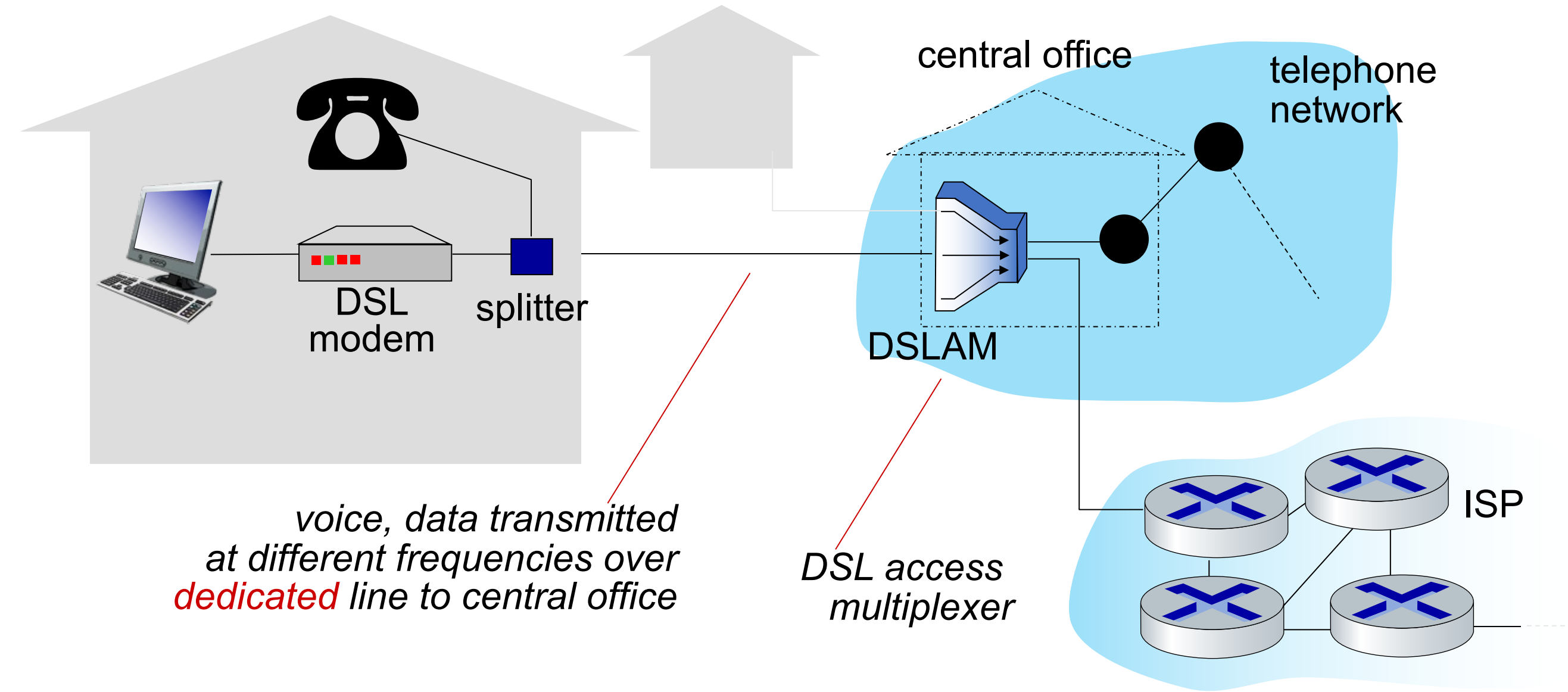
Example for shared network access: Cable



► Hybrid Fiber Coax

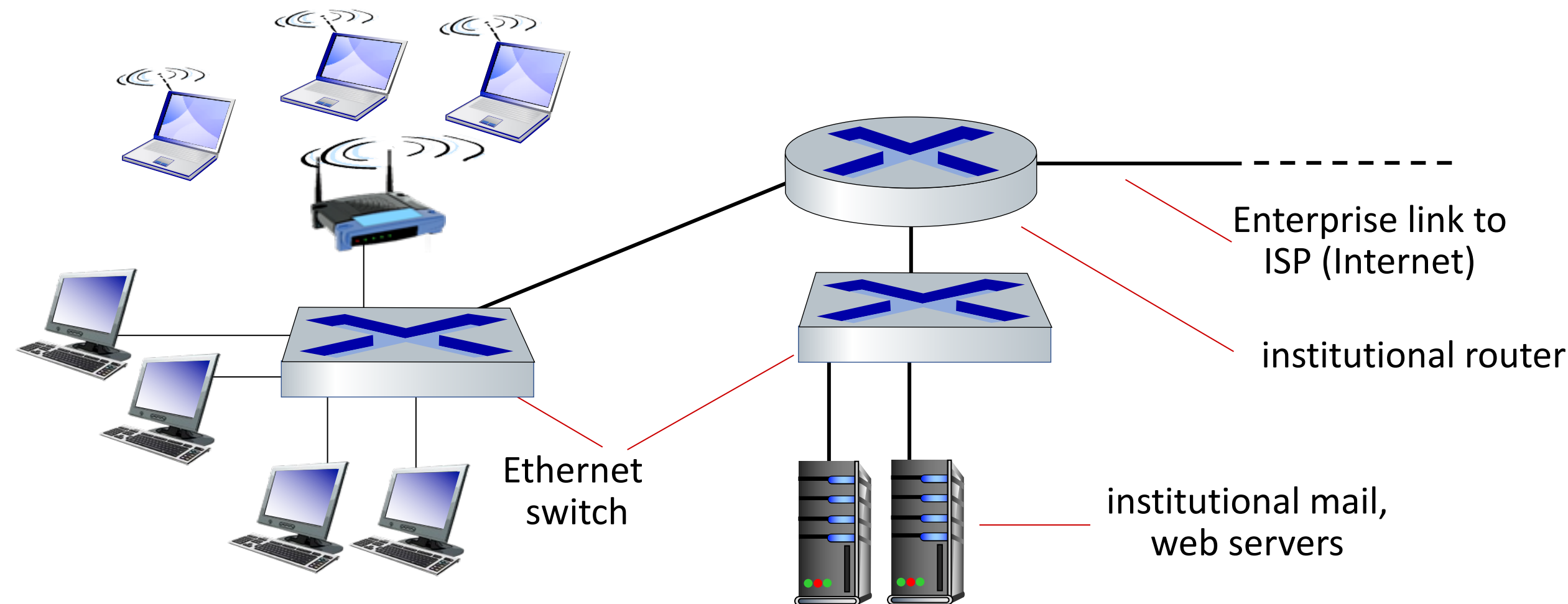
- Inherently asymmetric → downstream and upstream bandwidth differ
- Upstream to cable headend is shared by multiple access networks
- Data Over Cable Service Interface Specification (DOCSIS)
- Relevant standard, in Europe EuroDOCSIS

Example for dedicated network access: DSL



- ▶ Uses copper wires used in traditional landlines, signal is split
 - ▶ Data over DSL phone line goes to Internet
 - ▶ Voice over DSL phone line goes to telephone net

Specifics of enterprise networks

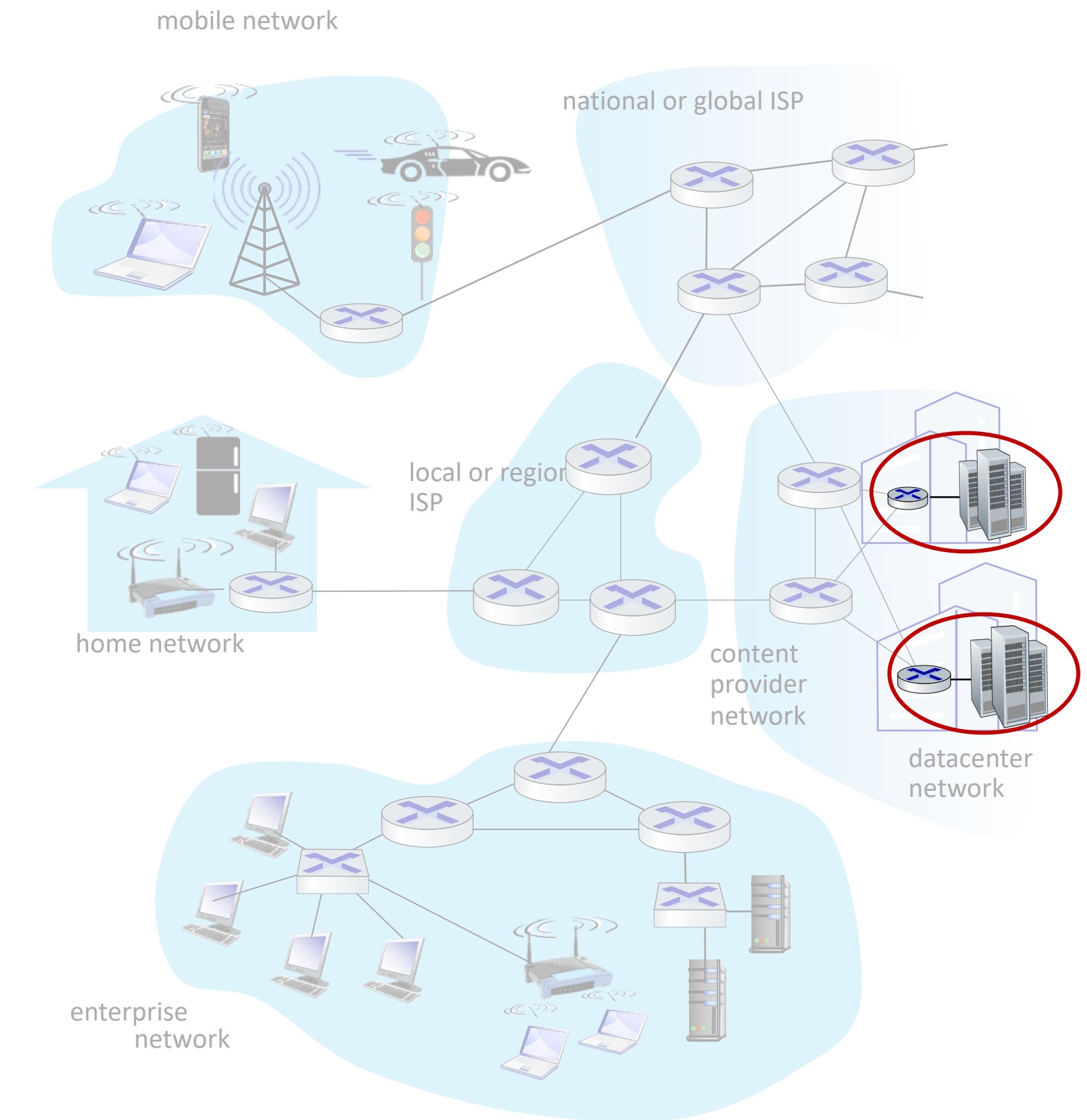


- ▶ Accommodate requirements of companies, universities, etc.
- ▶ More dedicated devices and usage of additional technologies, e.g.,
 - ▶ Virtual LANs
 - ▶ Power over ethernet
 - ▶ Client authenticated access
- ▶ Often different connections for backbone servers and client hosts

Specifics of data centre networks



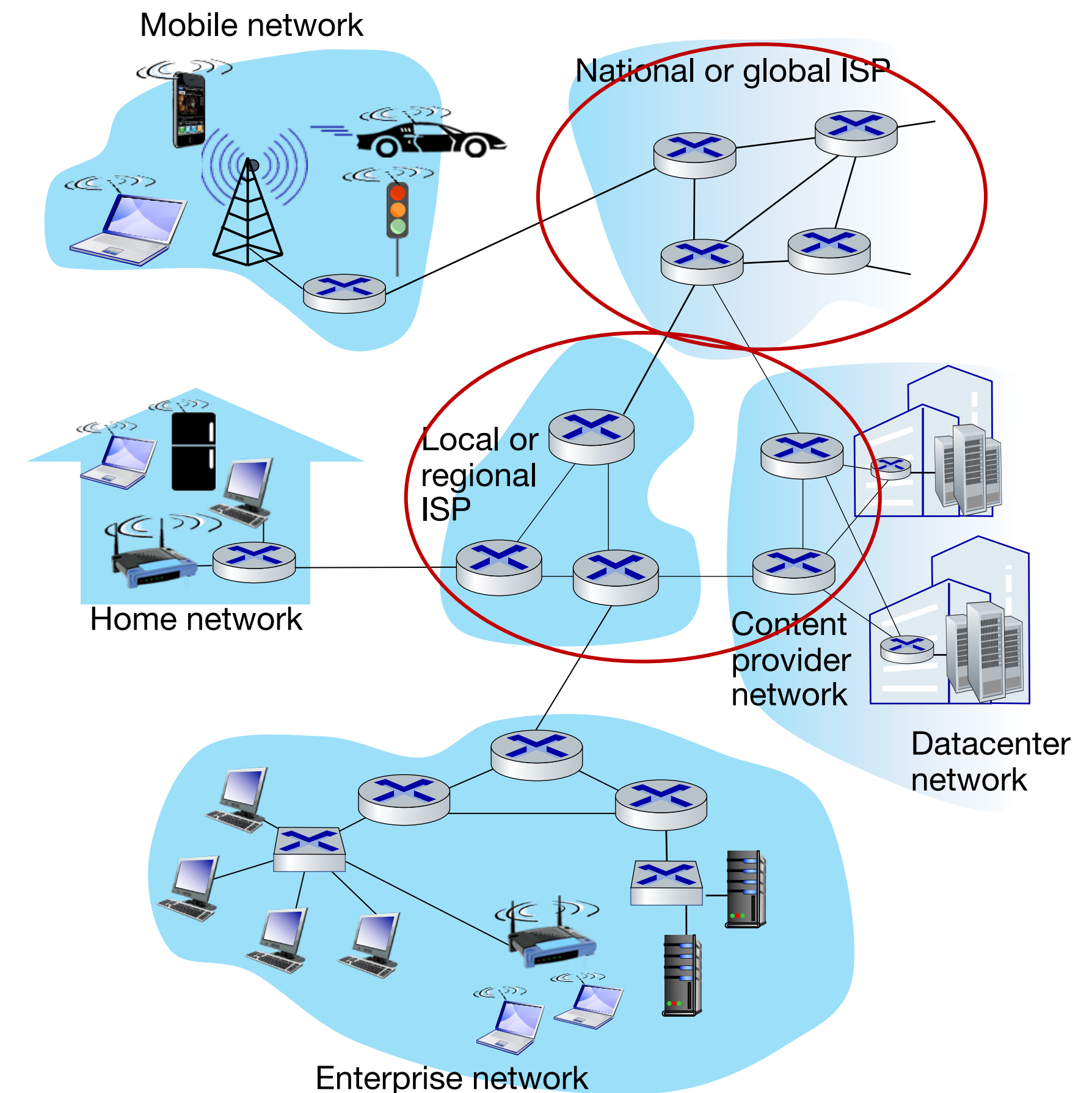
Courtesy: Massachusetts Green High Performance Computing Center (mghpcc.org)



- ▶ High bandwidth links are essential
- ▶ Mostly ethernet-based connections, up to currently 400G
- ▶ Depending on requirements specialised technologies, e.g., InfiBand

Network Topology

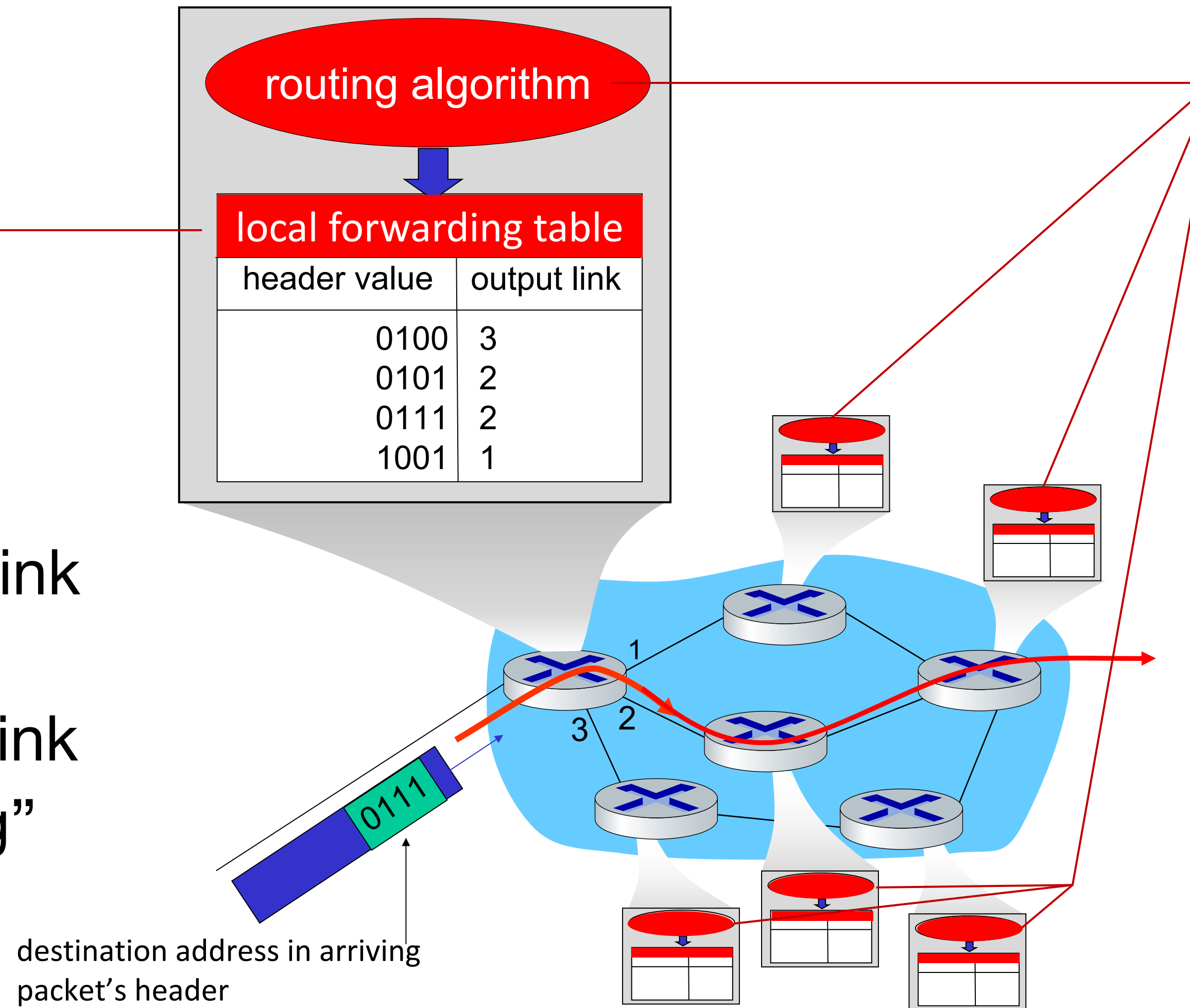
- ▶ The Network Edge
 - ▶ Hosts: clients and servers
 - ▶ Servers often in data centres
- ▶ Access Networks
 - ▶ Wired, wireless communication links
 - ▶ Provide connection to other networks through edge routers
- ▶ The Network Core
 - ▶ Mesh of interconnected routers
 - ▶ Network of networks
 - ▶ Key functions: Forwarding & Routing



Key functions: Forwarding & Routing

Forwarding:

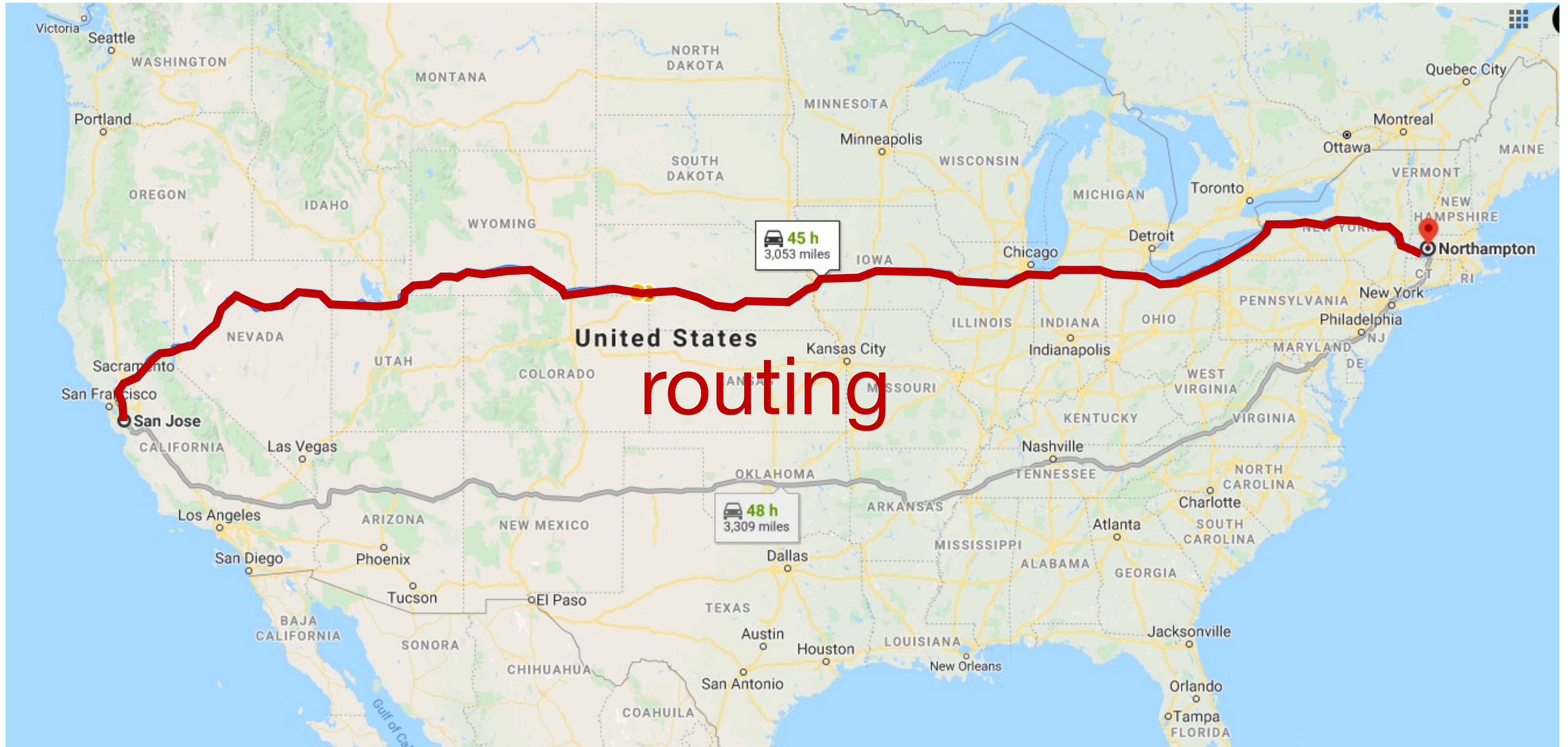
- ▶ *local* action:
move arriving packets from router's input link to appropriate router output link
aka "switching"



Routing:

- ▶ *global* action:
determine source-destination paths taken by packets
- ▶ Different routing algorithms

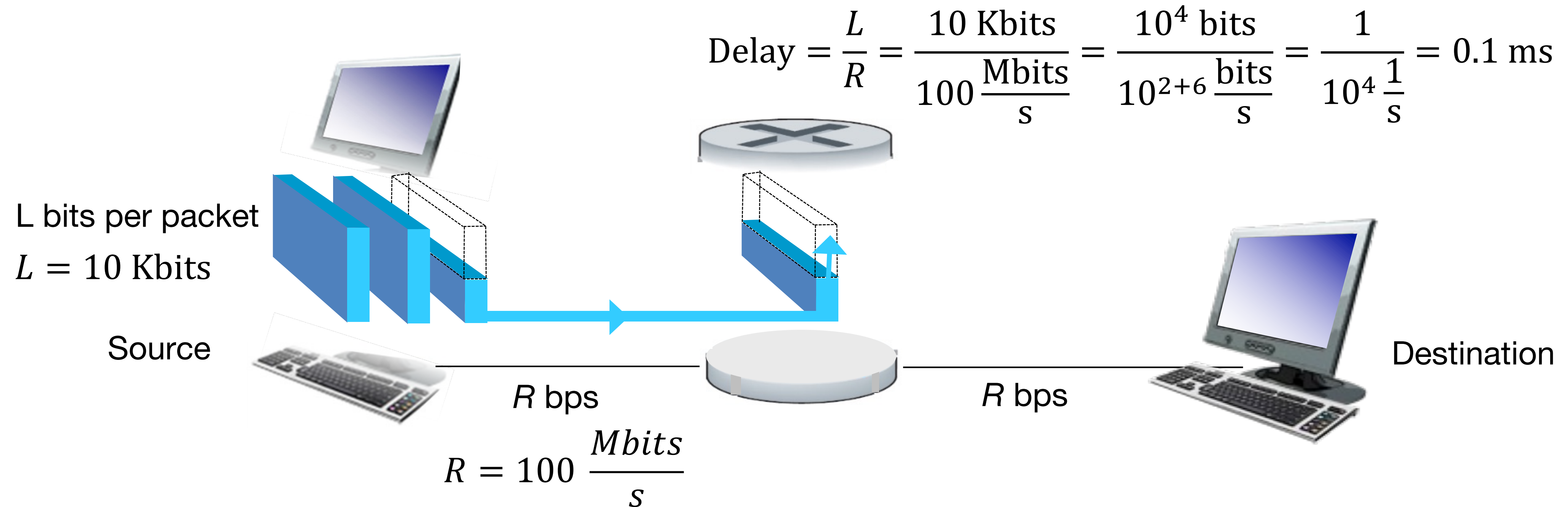
Key functions: Forwarding & Routing



Key functions: Forwarding & Routing

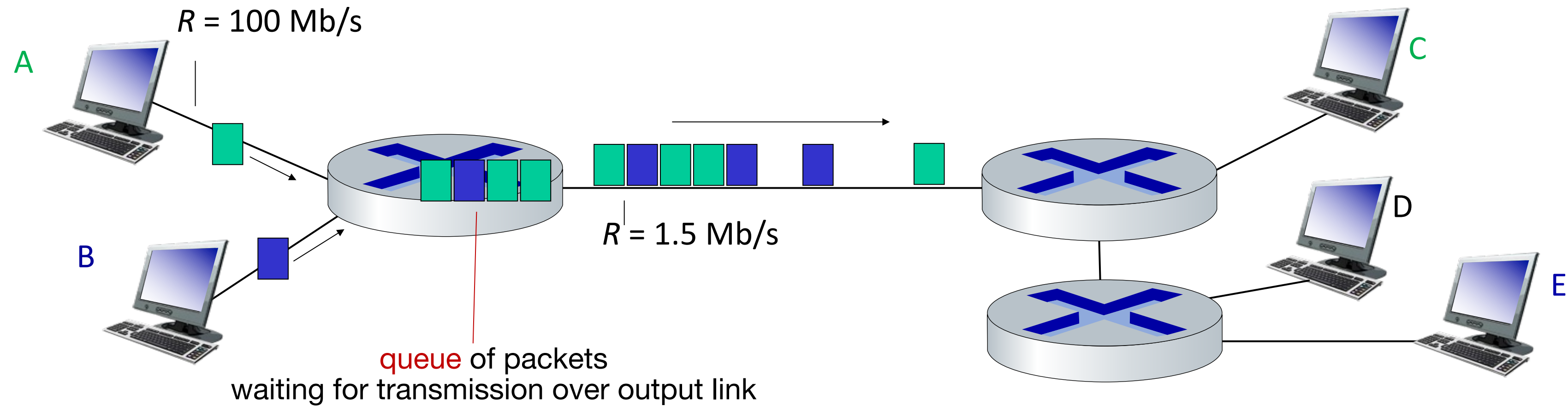


Packet Switching – Store and Forward



- **Packet Transmission Delay:** takes L/R seconds to transmit (push out) L-bit packet into link at R bps
- **Store and Forward:** entire packet must arrive at router before it can be transmitted on next link

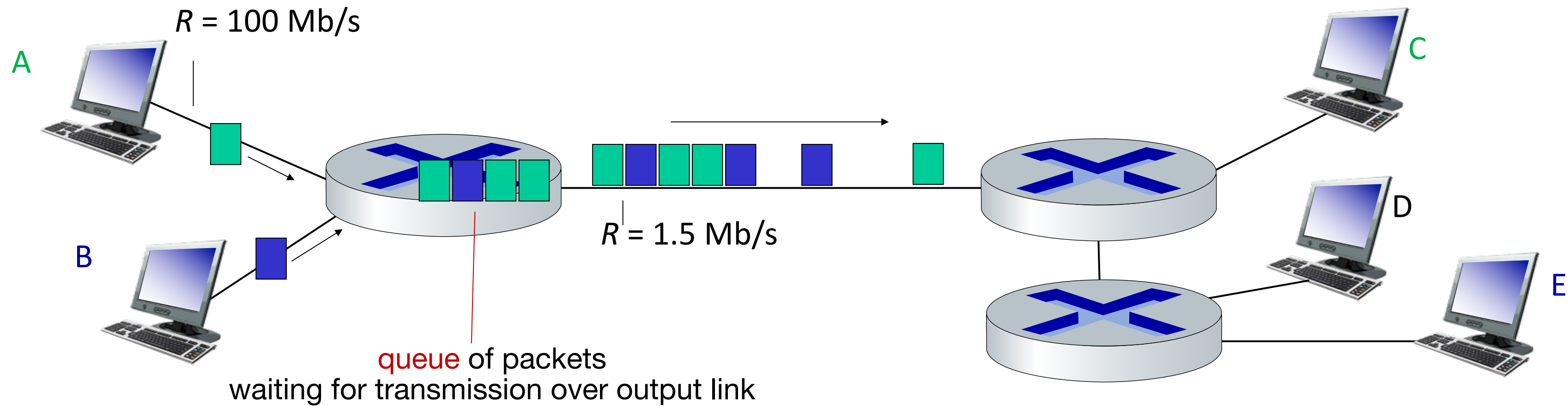
Packet Switching – Queueing



► **Queueing:** Occurs if work arrives faster than it can be served

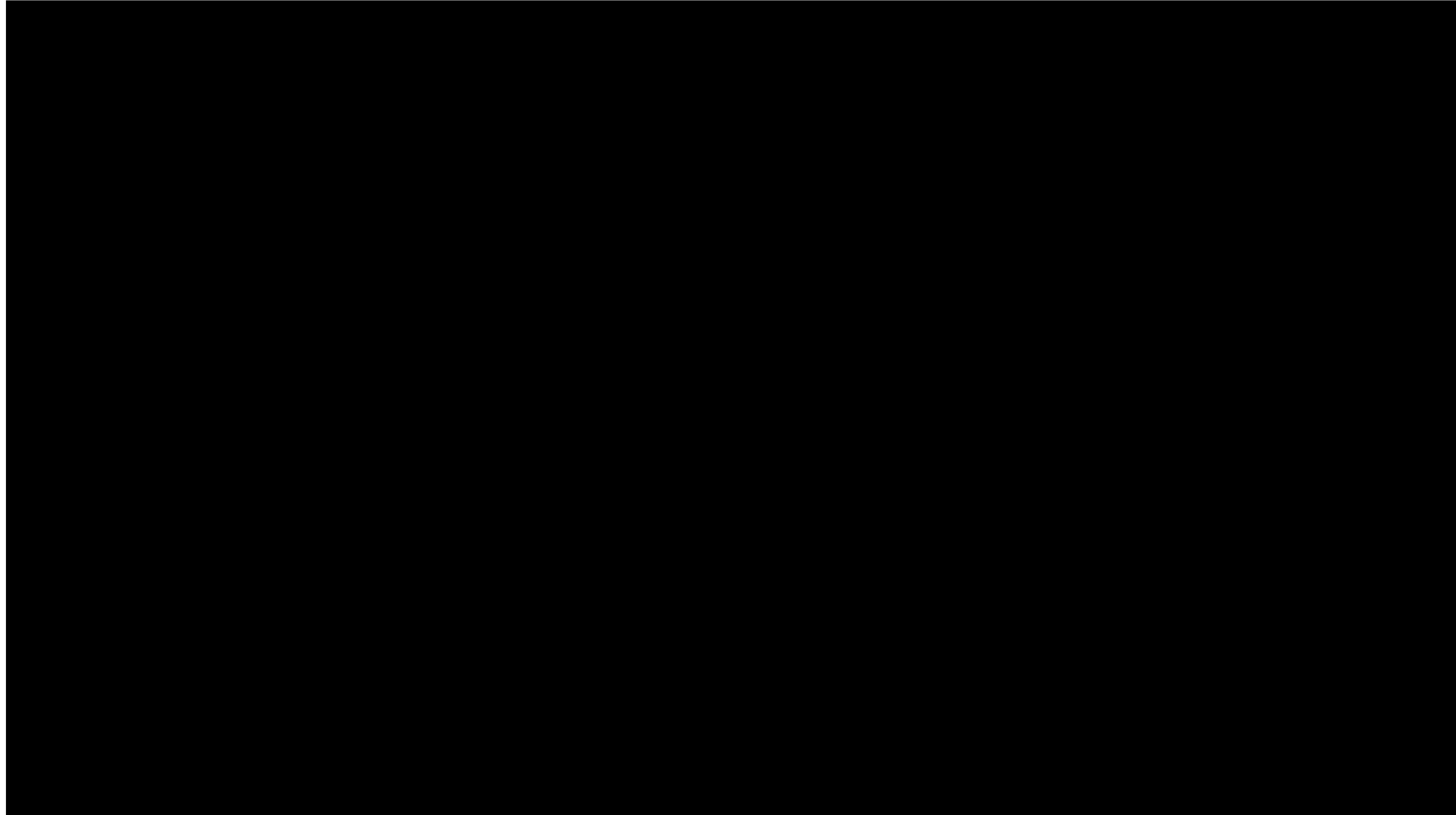


Packet Switching – Queueing



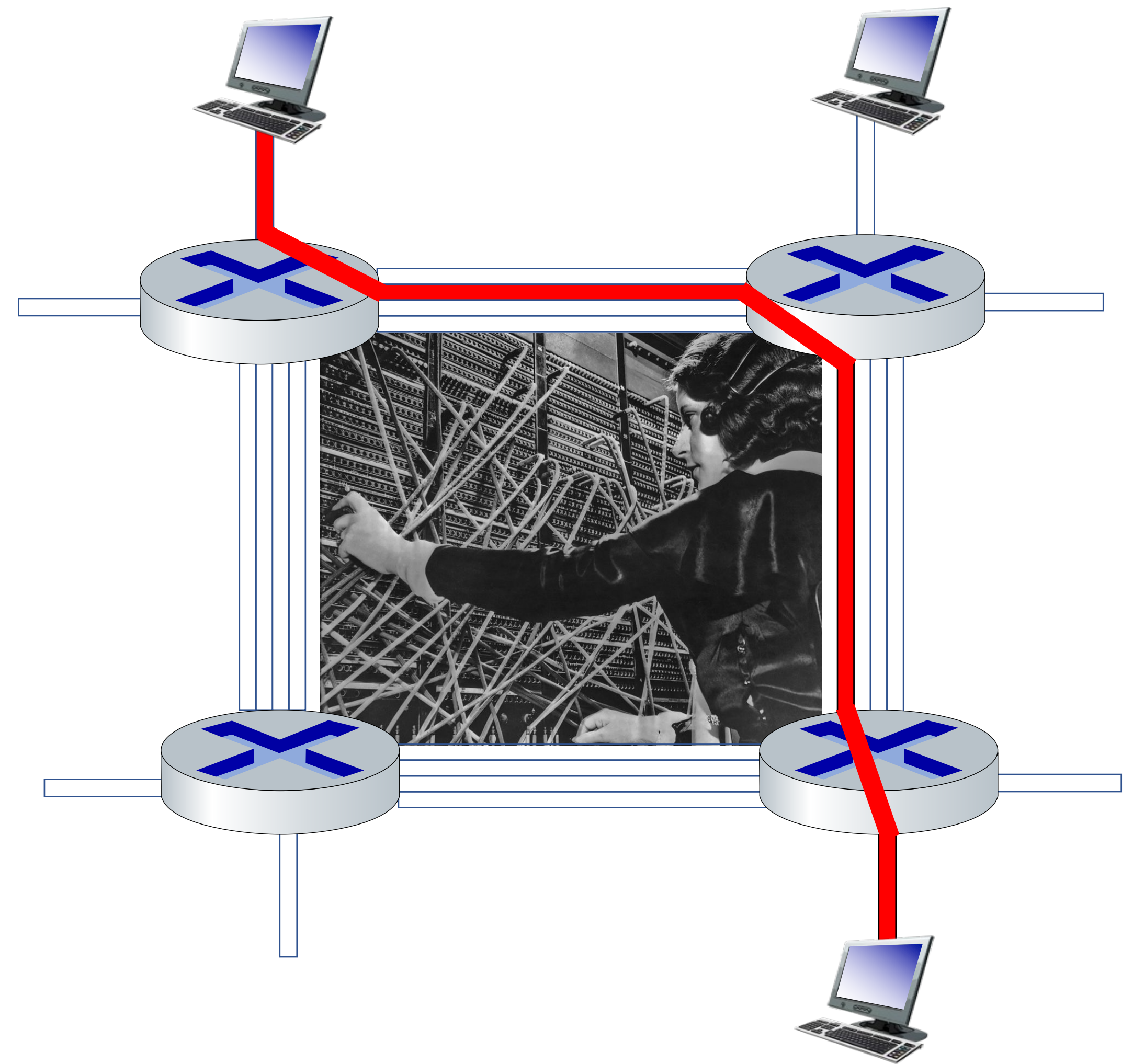
- ▶ **Queueing:** Occurs if work arrives faster than it can be served
- ▶ **Packet loss:** If arrival rate exceeds transmission rate
 - ▶ First packets will queue as long as the memory of the router allows it
 - ▶ Then packets will be dropped (lost)

~~Life Lesson~~ ~~Interlude:~~ Why one queue is better than multiple



Alternative to Packet Switching: Circuit Switching

- ▶ Requires exclusive allocation of end-to-end resource
- ▶ Segments remain idle if bandwidth not fully used
- ▶ Example connection
 - ▶ Each link has four circuits
 - ▶ One circuit has to be allocated for each link allocated
 - ▶ Max. four connections per link
- ▶ Commonly used in traditional telephone lines

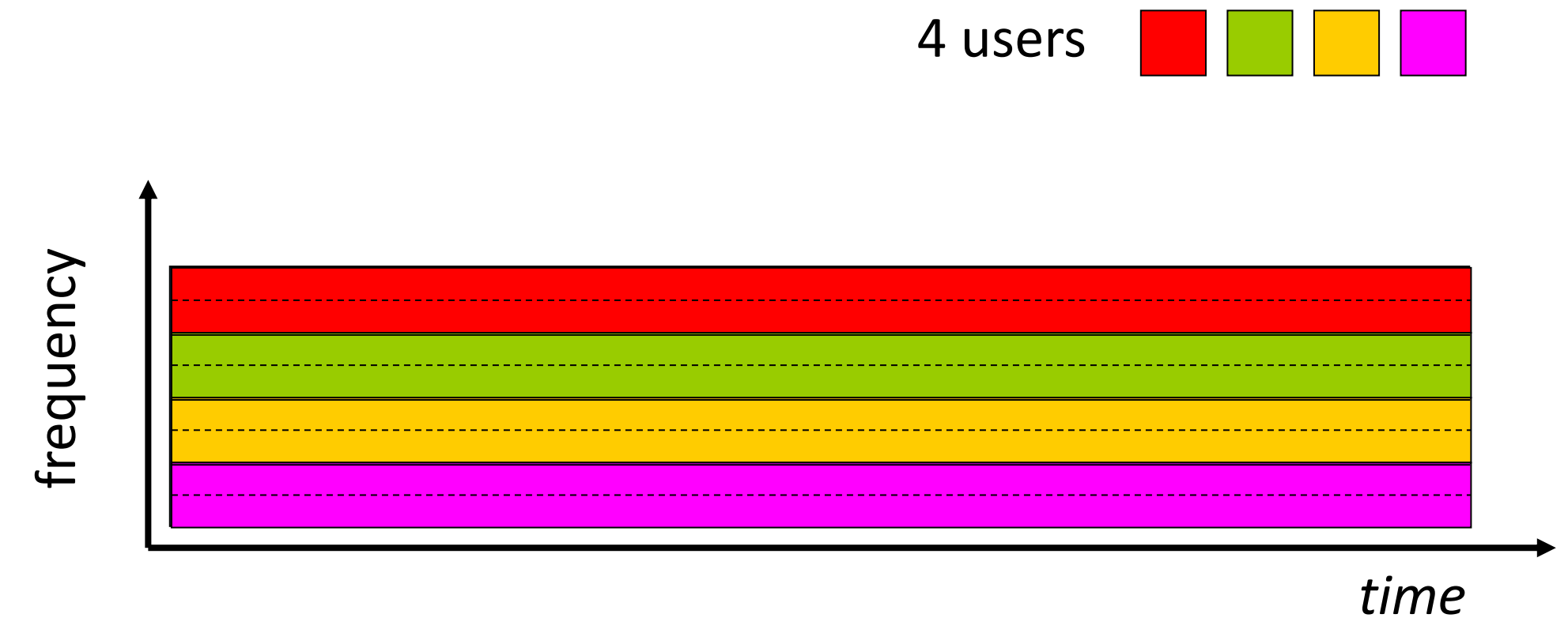


Adapted from slides by J.F. Kurose and K.W. Ross

Alternative to Packet Switching: Circuit Switching

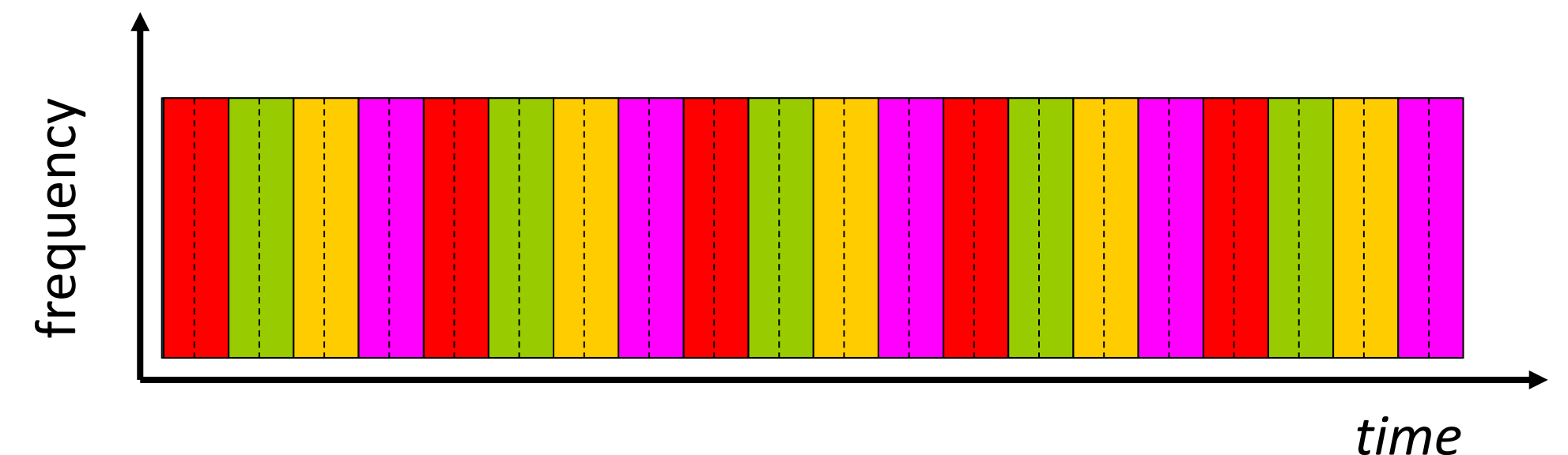
► Frequency Division Multiplexing (FDM)

- Optical, electromagnetic frequencies divided into (narrow) bands
- Each connection allocated its own band, can transmit at max. rate of that band



► Time Division Multiplexing (TDM)

- Time divided into slots
- Each call allocated periodic slots, can transmit at maximum rate of wider band
- Transmission only during allocated slots

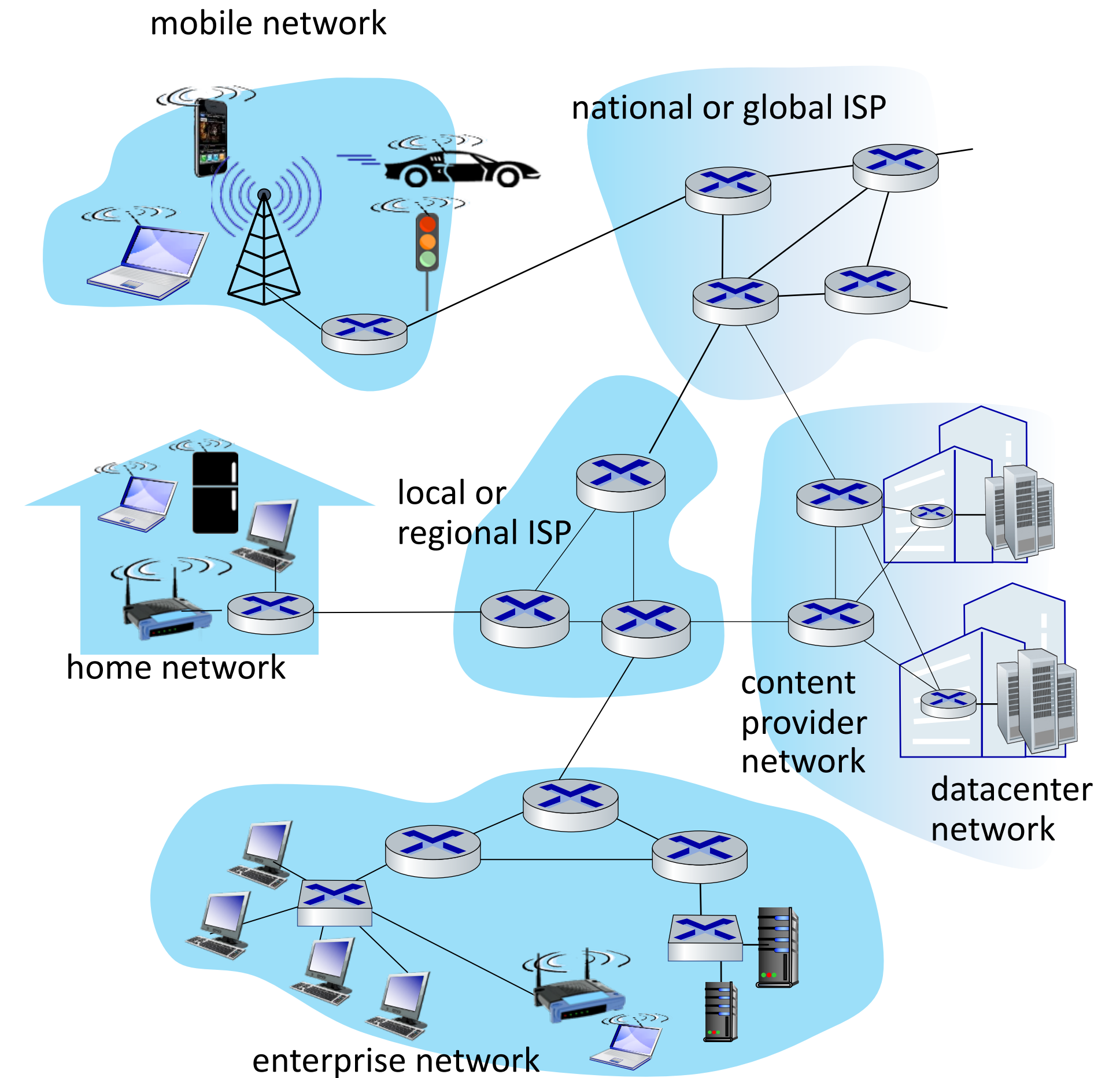


Packet Switching Circuit Switching

- ▶ Packet switching is more flexible and great for “bursty” data
 - ▶ Resource can be shared when not needed
 - ▶ Simpler, no call setup
- ▶ Circuit Switching is better to deliver reliable data transfer
 - ▶ No packet loss due to congestion
- ▶ Is it possible to provide circuit-like behaviour with packet switching?
 - ▶ Kinda, QoS techniques exist

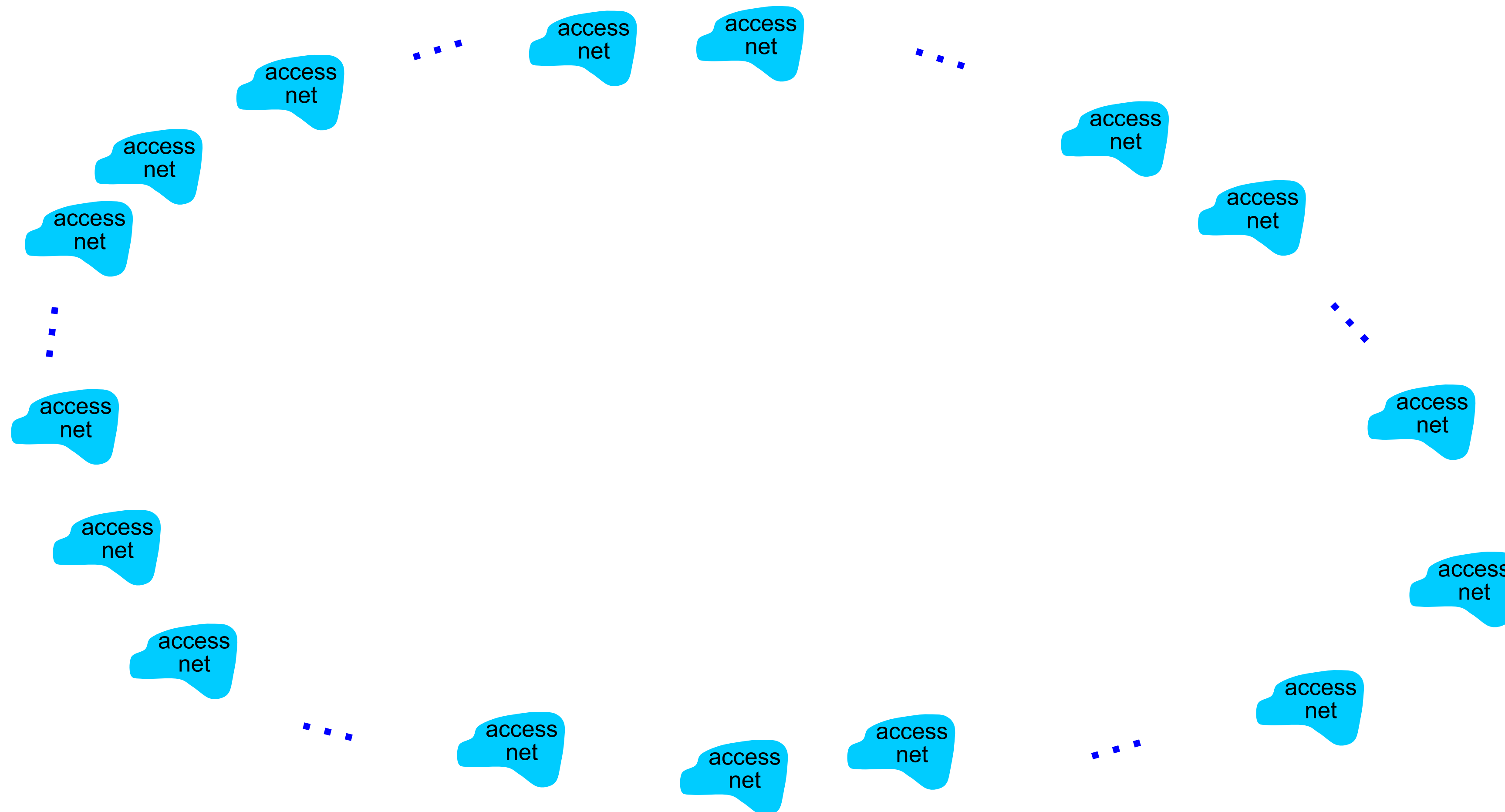
The Internet: A network of networks

- ▶ Hosts connect to Internet via ISPs
- ▶ But how are these ISPs interconnected? 🤔
- ▶ Required so anyone can connect to anyone else
- ▶ It's a network of networks



The Internet Topology: A network of networks

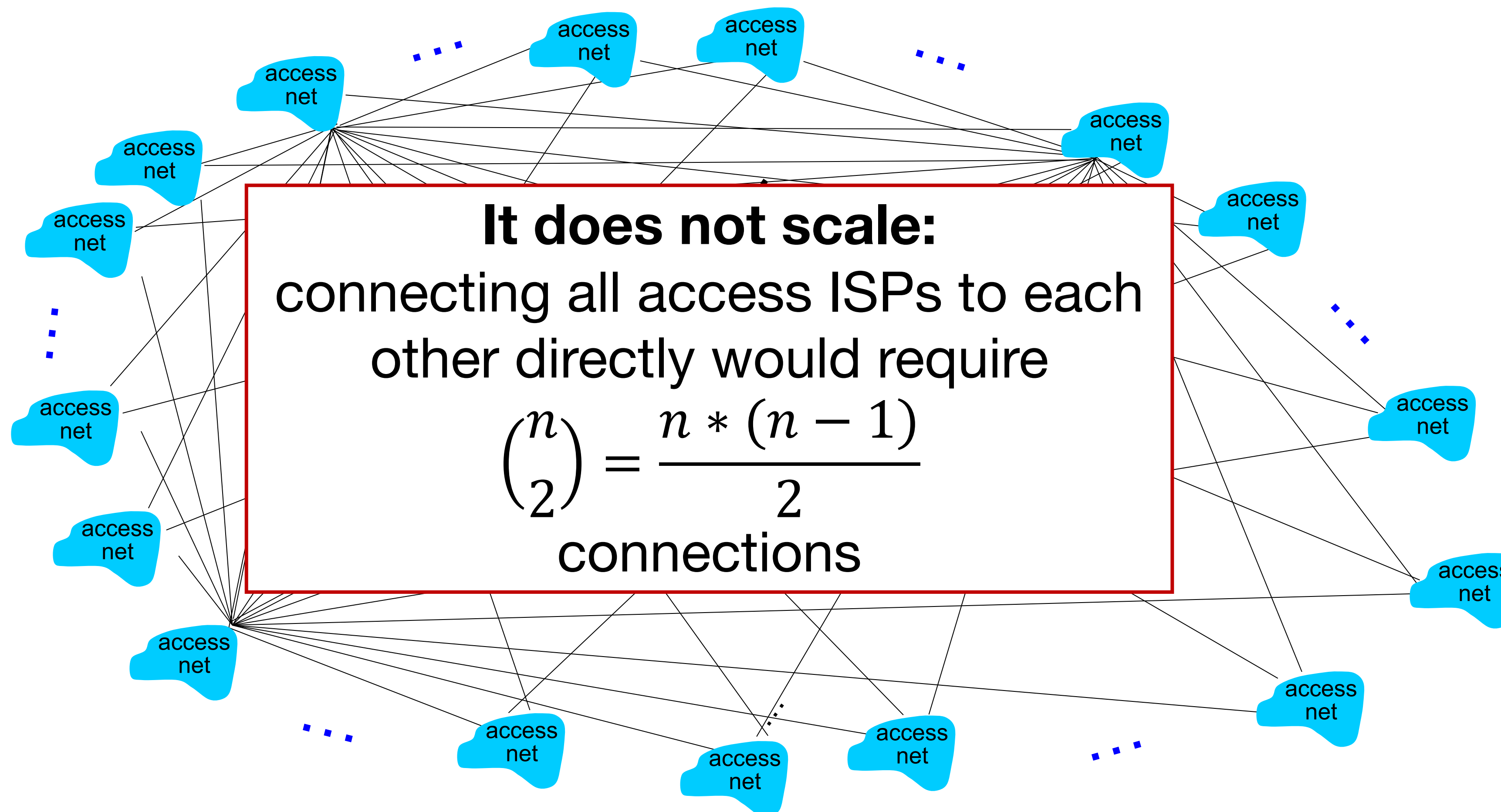
► Given the number of networks to interconnect → how to best do it?



Adapted from slides by J.F Kurose and K.W. Ross

The Internet Topology: A network of networks

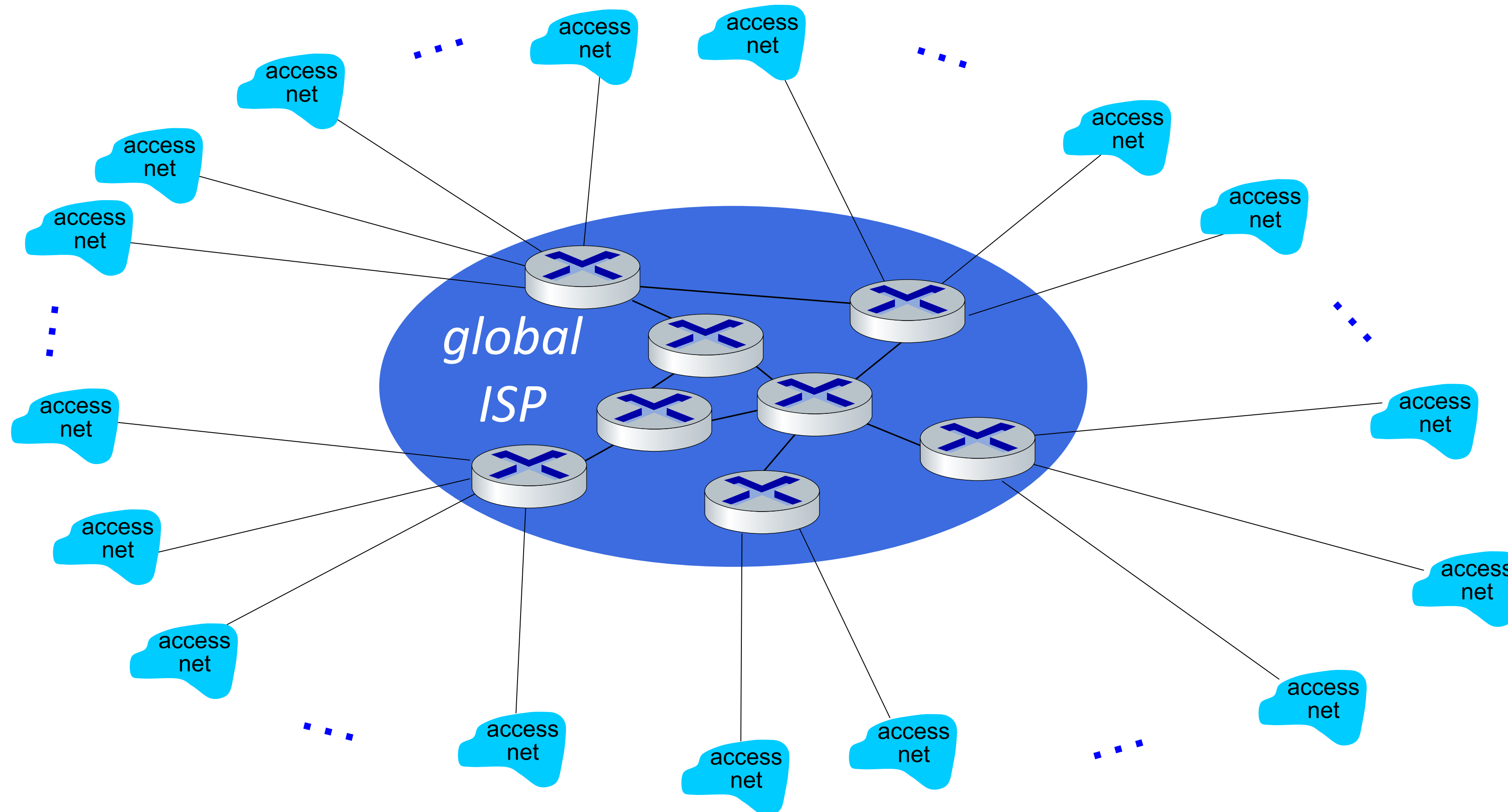
- Given the number of networks to interconnect → how to best do it?



Adapted from slides by J.F Kurose and K.W. Ross

The Internet Topology: A network of networks

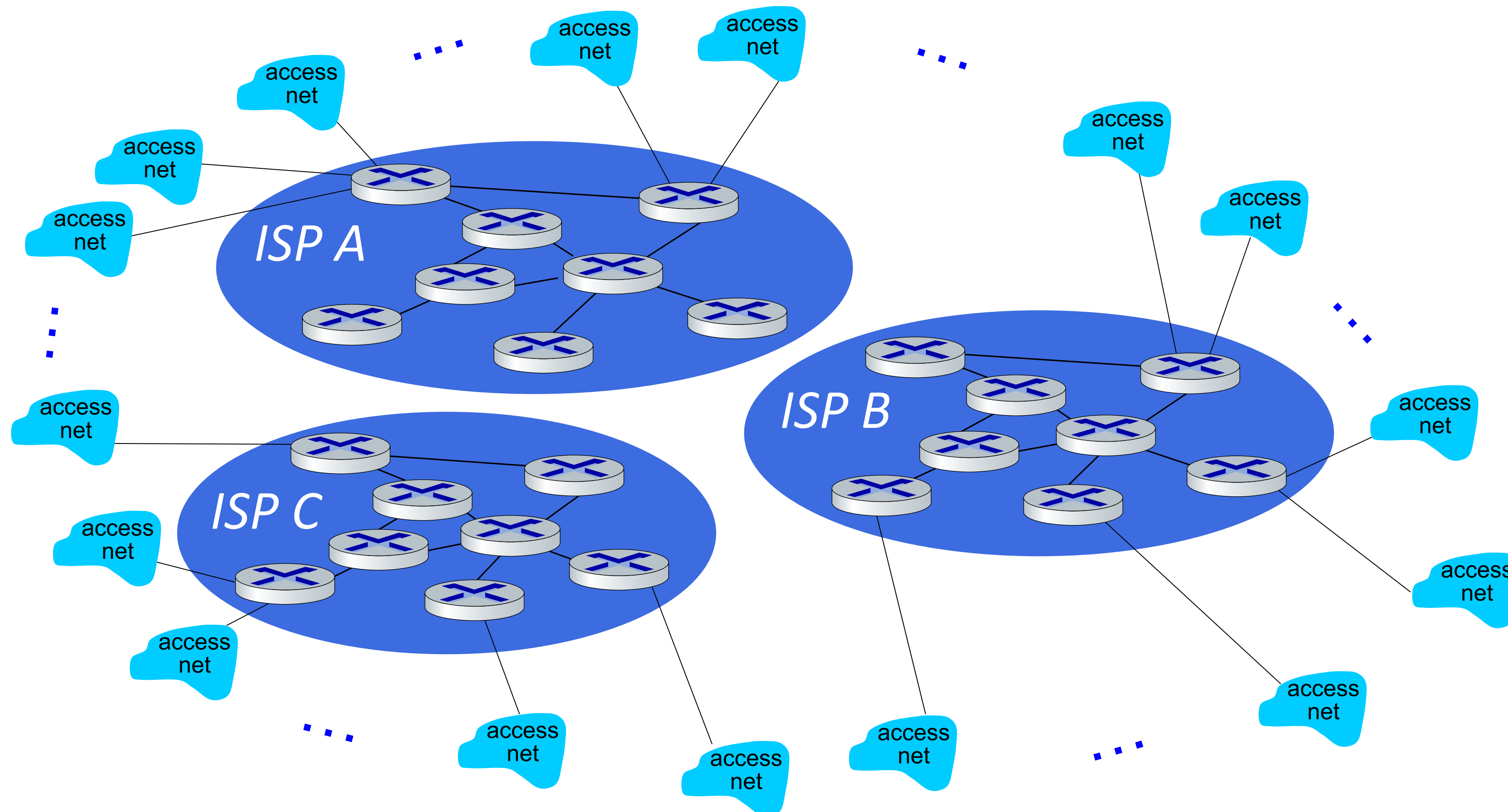
- ▶ Given the number of networks to interconnect → how to best do it?
- ▶ Provide tier 1 ISP that connects everyone



Adapted from slides by J.F. Kurose and K.W. Ross

The Internet Topology: A network of networks

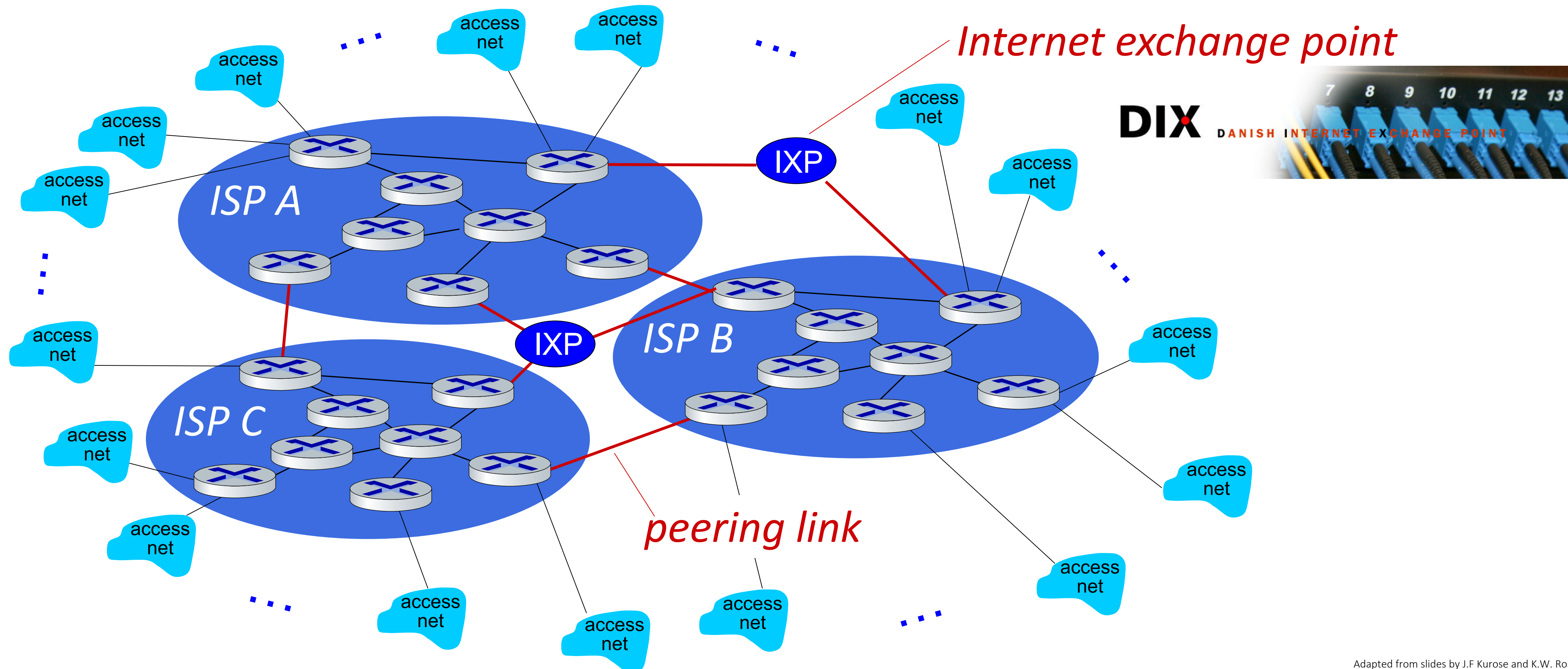
- ▶ Provide tier 1 ISP that connects everyone
- ▶ But if a viable business there will be multiple



Adapted from slides by J.F Kurose and K.W. Ross

The Internet Topology: A network of networks

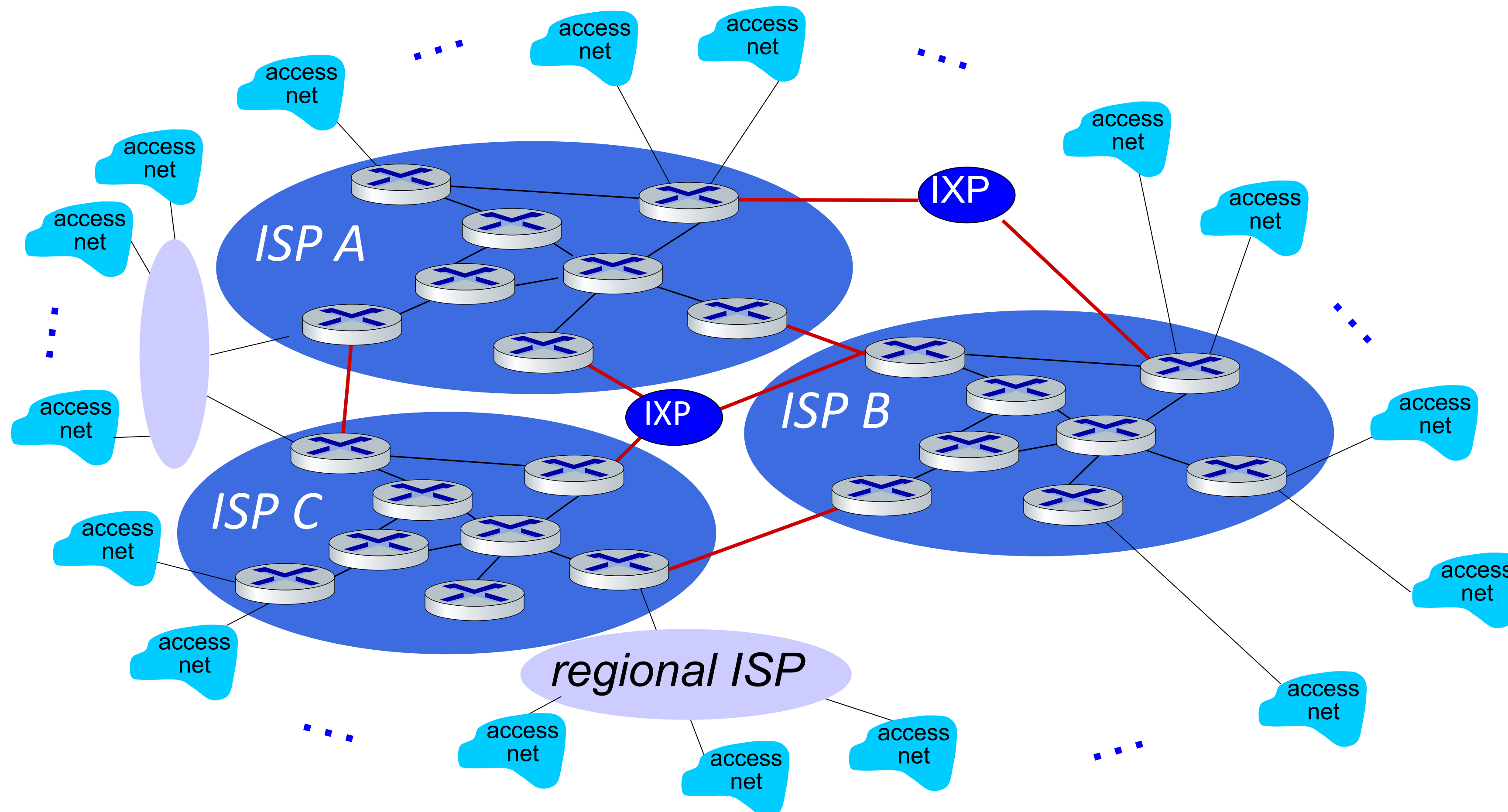
- Provide tier 1 ISP that connects everyone
- But if a viable business there will be multiple and with need to be connected



Adapted from slides by J.F Kurose and K.W. Ross

The Internet Topology: A network of networks

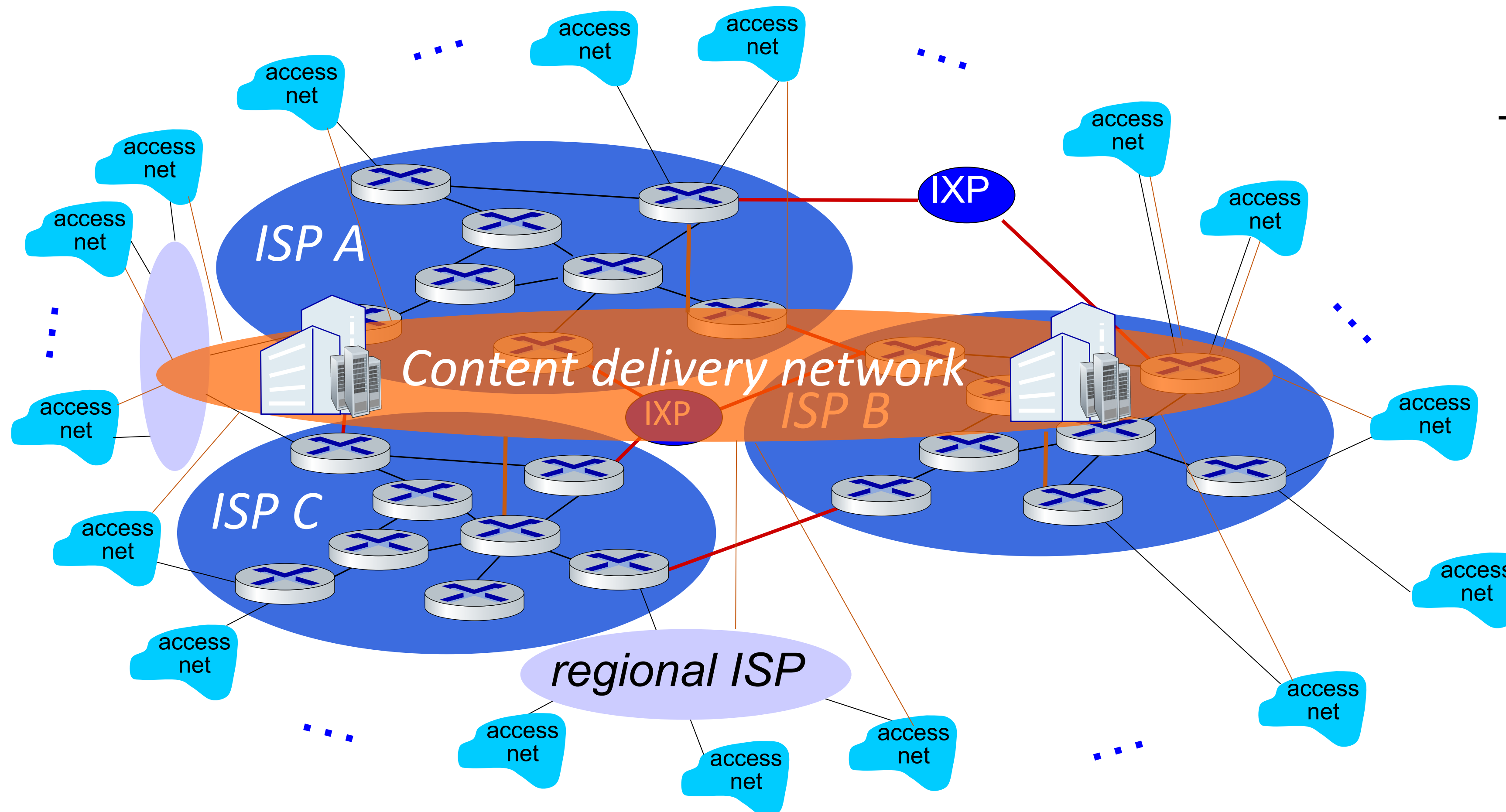
- ▶ Provide tier 1 ISPs that are interconnected
- ▶ Regional networks may arise to connect access ISPs and tier 1 ISPs



Adapted from slides by J.F Kurose and K.W. Ross

The Internet Topology: A network of networks

- Provide tier 1 ISPs that are interconnected and connect to Regional networks
- Content Providers (e.g., Google, Akamai) might have nodes in all networks

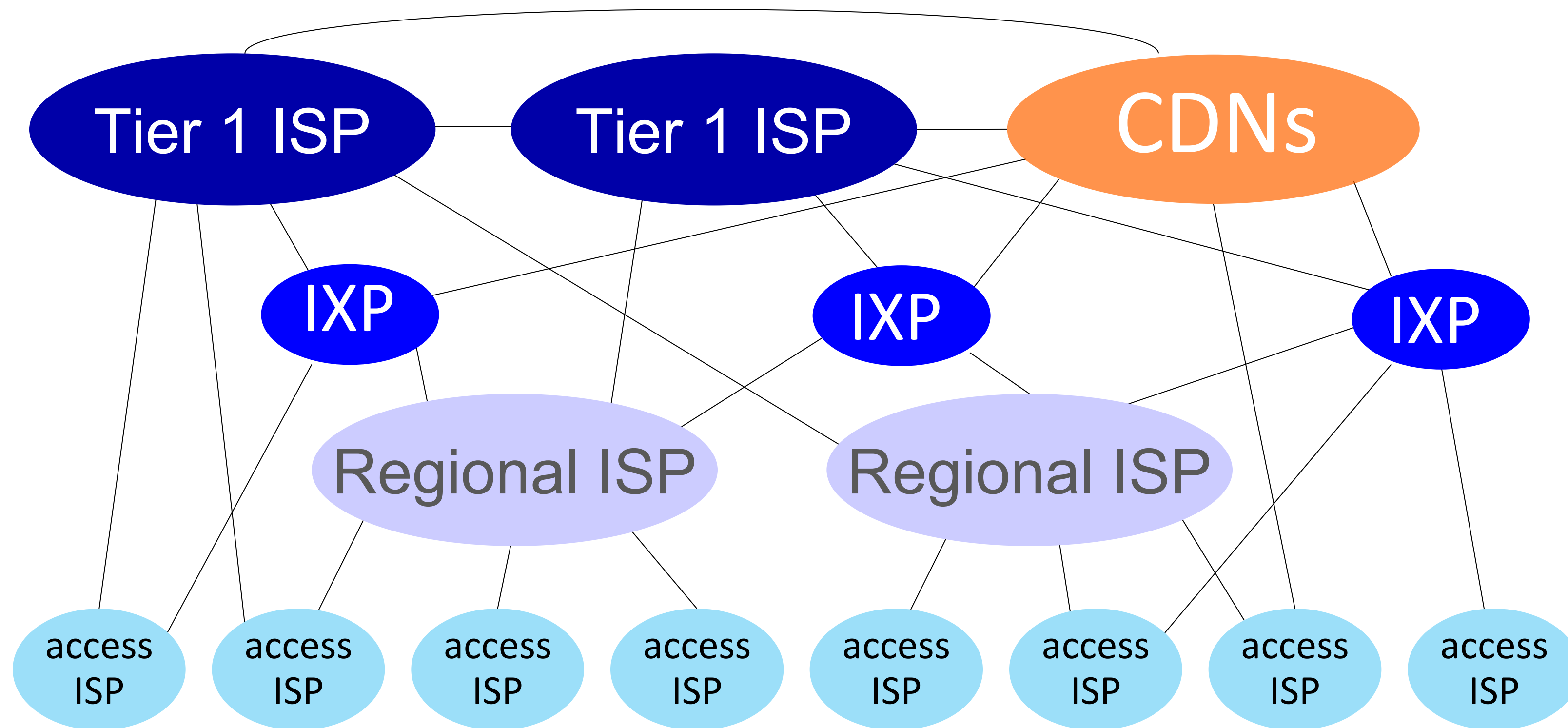


Q: Why?

To be close to the customer

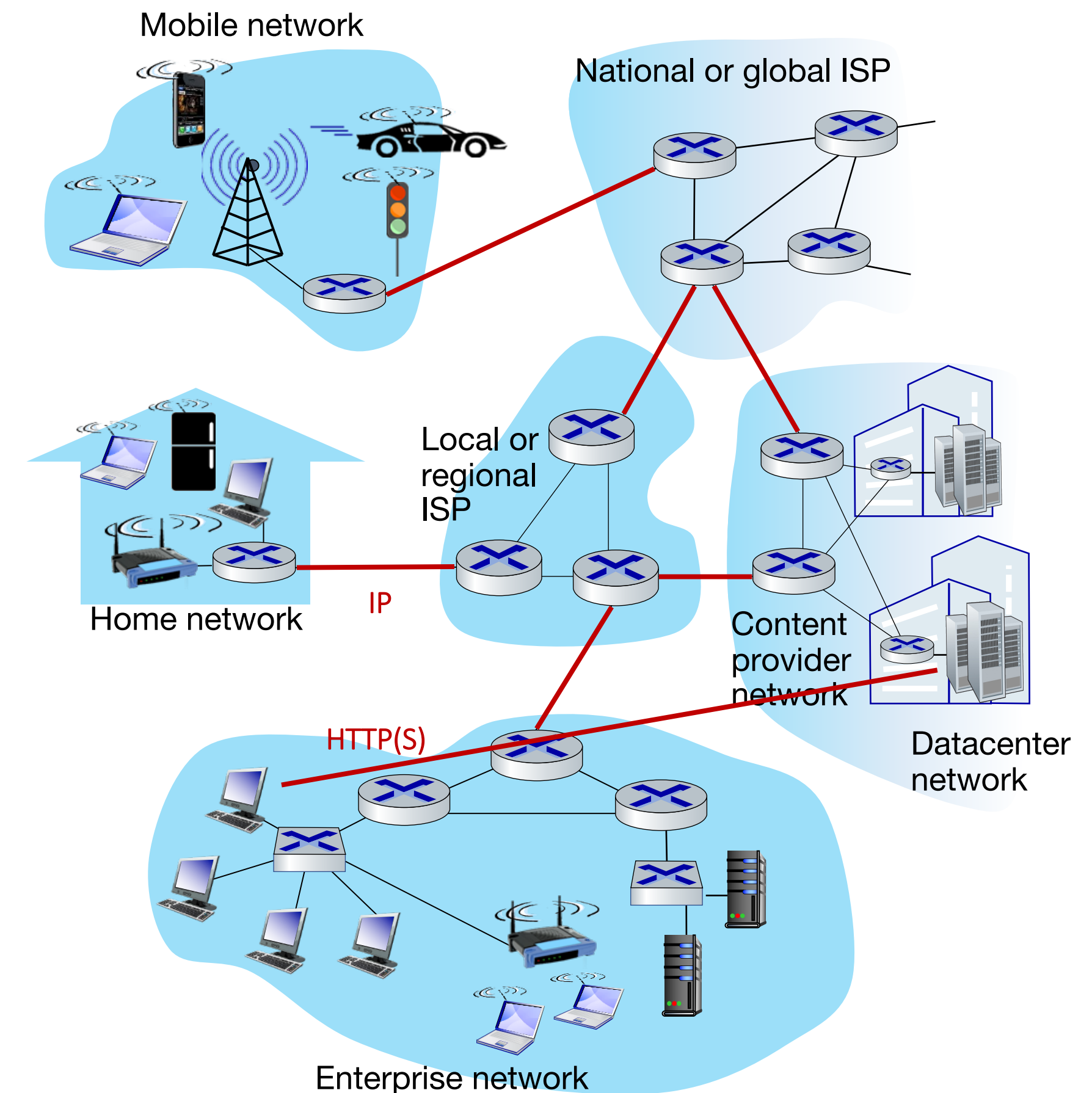
The Internet Topology: A hierarchy of networks

- ▶ Tier 1 core: Large ISPs and Content Provider Networks
- ▶ Interconnection facilitated by IXPs and Regional ISPs
- ▶ Tier 2 core: Access ISPs



The Internet Topology

- ▶ Internet
 - ▶ Network of Networks
 - ▶ Interconnected ISPs (Internet Service Providers)
- ▶ Protocols
 - ▶ The language of computer networks
 - ▶ Control sending & receiving of messages
- ▶ Internet Standards
 - ▶ IETF: Internet Engineering Task Force
 - ▶ RFC: Request for Comments
- ▶ Local Area Network Standards
 - ▶ IEEE: Institute of Electrical and Electronics Engineers
 - ▶ Standards Family: IEEE 802



Learning Goal BD70D1

What is a Network Protocol?

Textbook Definition

A protocol defines the format and the order of messages sent and received among network entities, as well as the actions taken on message transmission and/or receipt.

- J.F. Kurose & K.W. Ross, Computer Networking

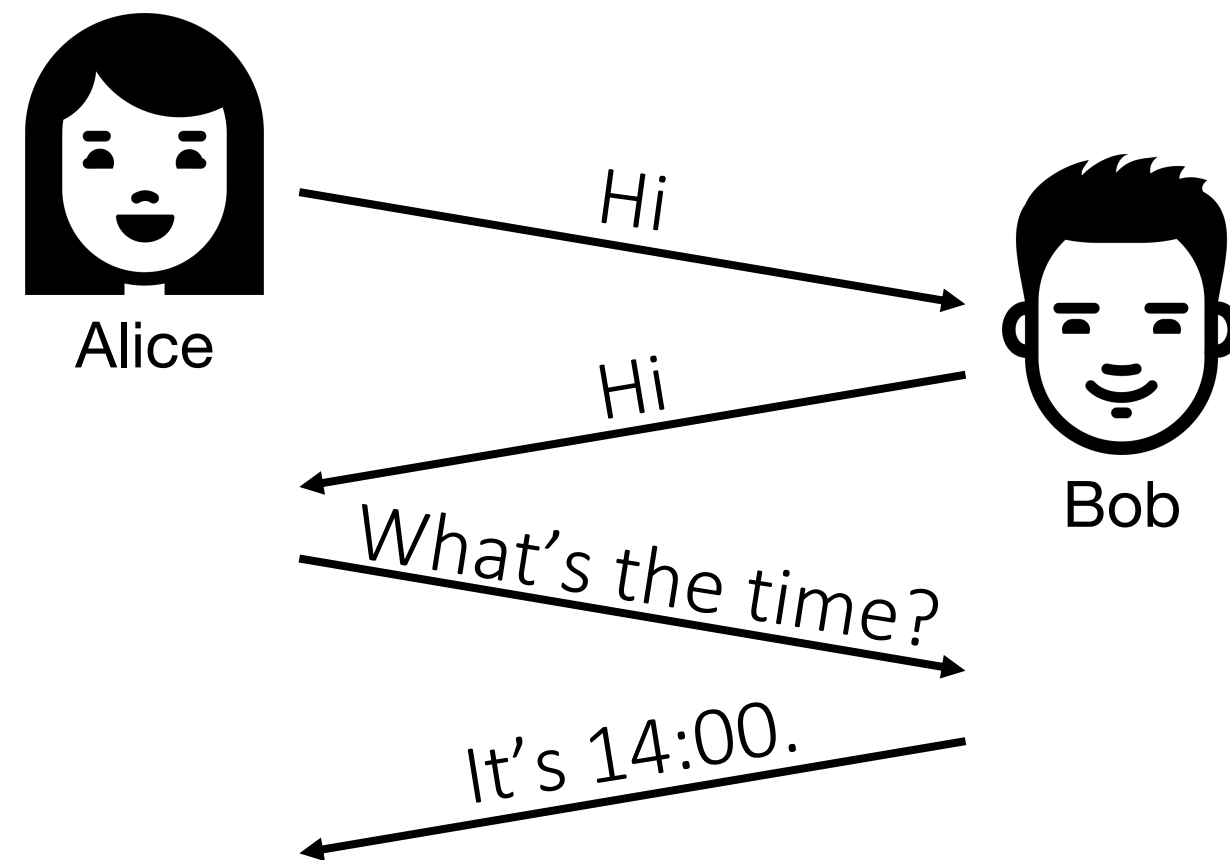
- ▶ They specify rules for
 - ▶ How to format messages
 - ▶ Which actions should be taken when messages are received or other events take place

- ▶ We do this as humans as well

Intuitive Understanding

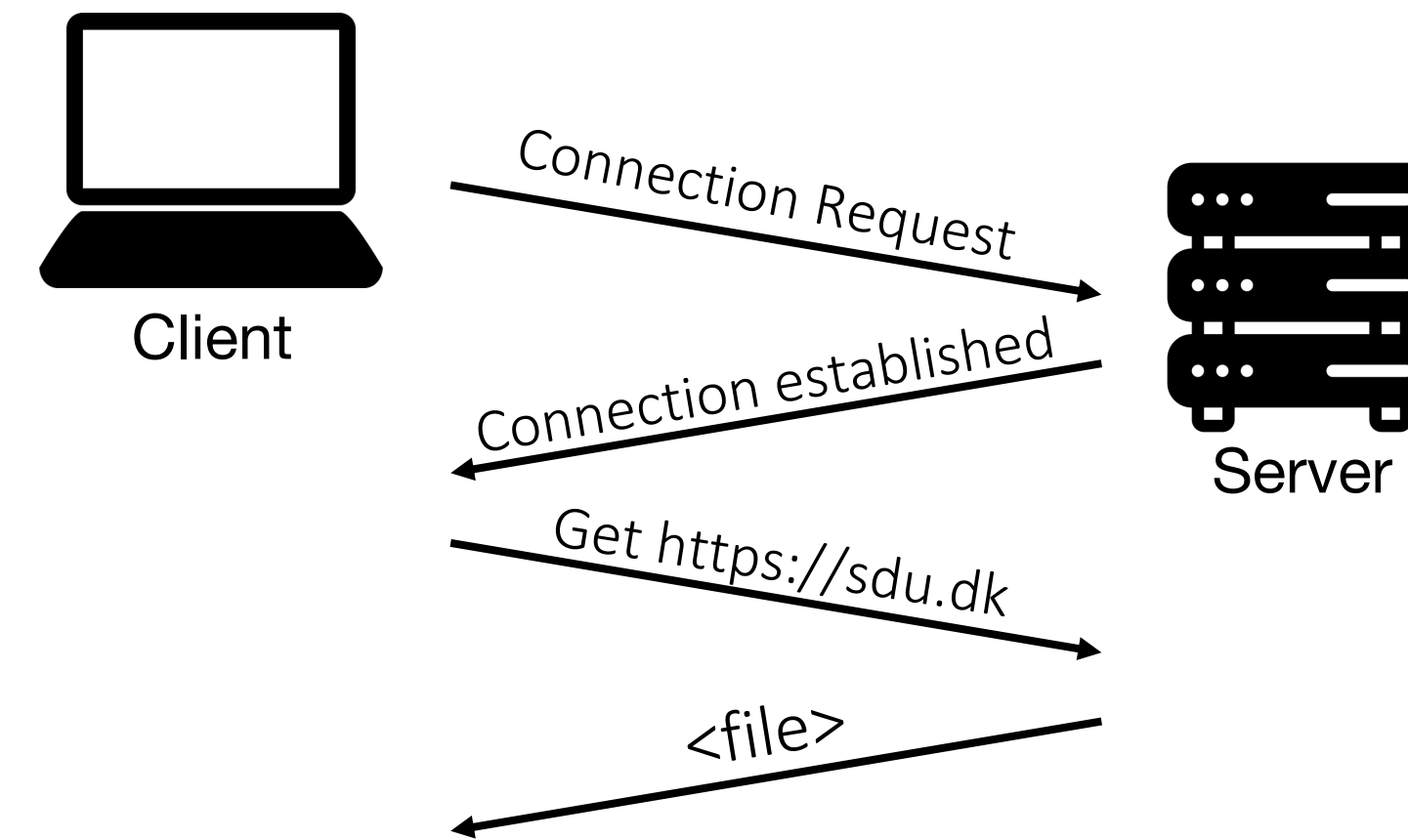
- ▶ They specify rules for
 - ▶ How to format messages
 - ▶ Which actions should be taken when messages are received or other events take place
- ▶ Example

Human protocol



Time

Network protocol



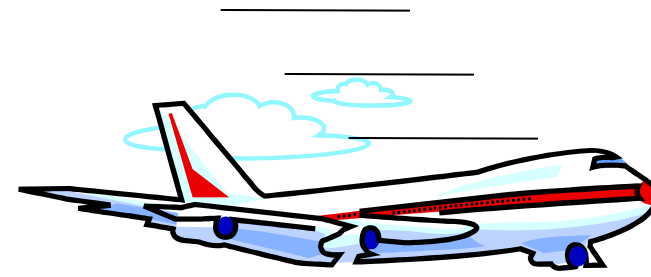
Learning Goal 10C7F8

How do Network Protocols interact?

Protocol Layers

- ▶ Networks are complex with many pieces and many different protocols
 - ▶ Hosts
 - ▶ Routers
 - ▶ HTTP
 - ▶ DNS
 - ▶ ...
- ▶ How can we best organise this mess of things a.k.a. our network?
- ▶ Different layers take care of specific tasks and hand-off to the next
- ▶ Example: Air Travel

Example: Air Travel



————— *end-to-end transfer of person plus baggage* —————→

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

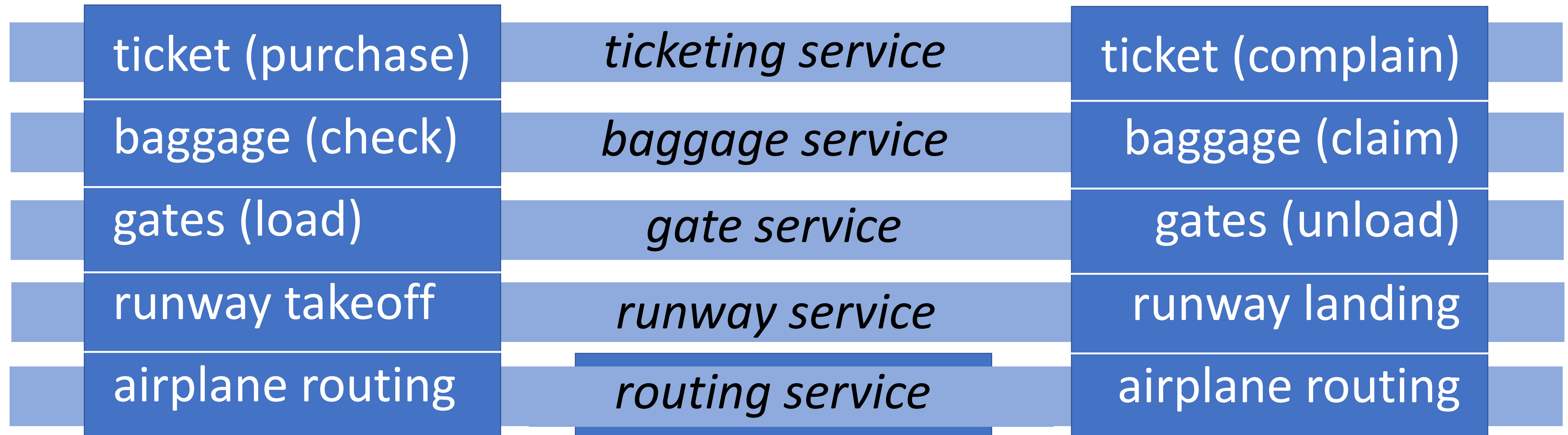
runway landing

airplane routing

airplane routing

- ▶ How to organise the system of airline travel with its many steps?
- ▶ How about as layers!

Example: Air Travel



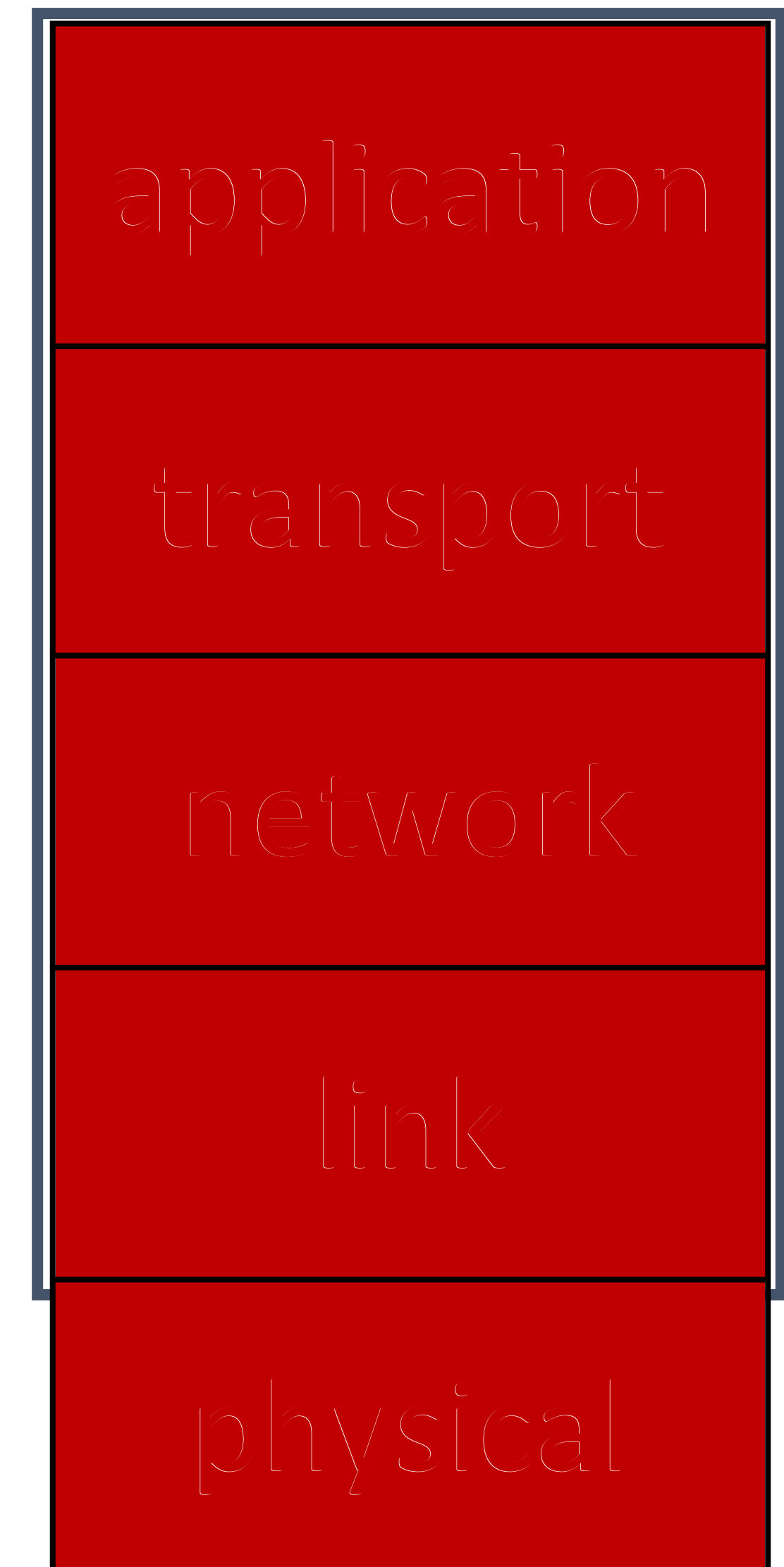
- ▶ Each layer implements a service
- ▶ Relying on services provided by layer below

Why is layering a good idea?

- ▶ Approach to model, design, and discuss complex systems
- ▶ Explicit structure allows identification and description of relationships
 - ▶ Reference model with abstraction
- ▶ Modularisation eases maintenance and updating of the system
 - ▶ Layers can change, since other layers are unaffected if same service/interface
 - ▶ E.g., change in gate does not affect other layers in air travel example

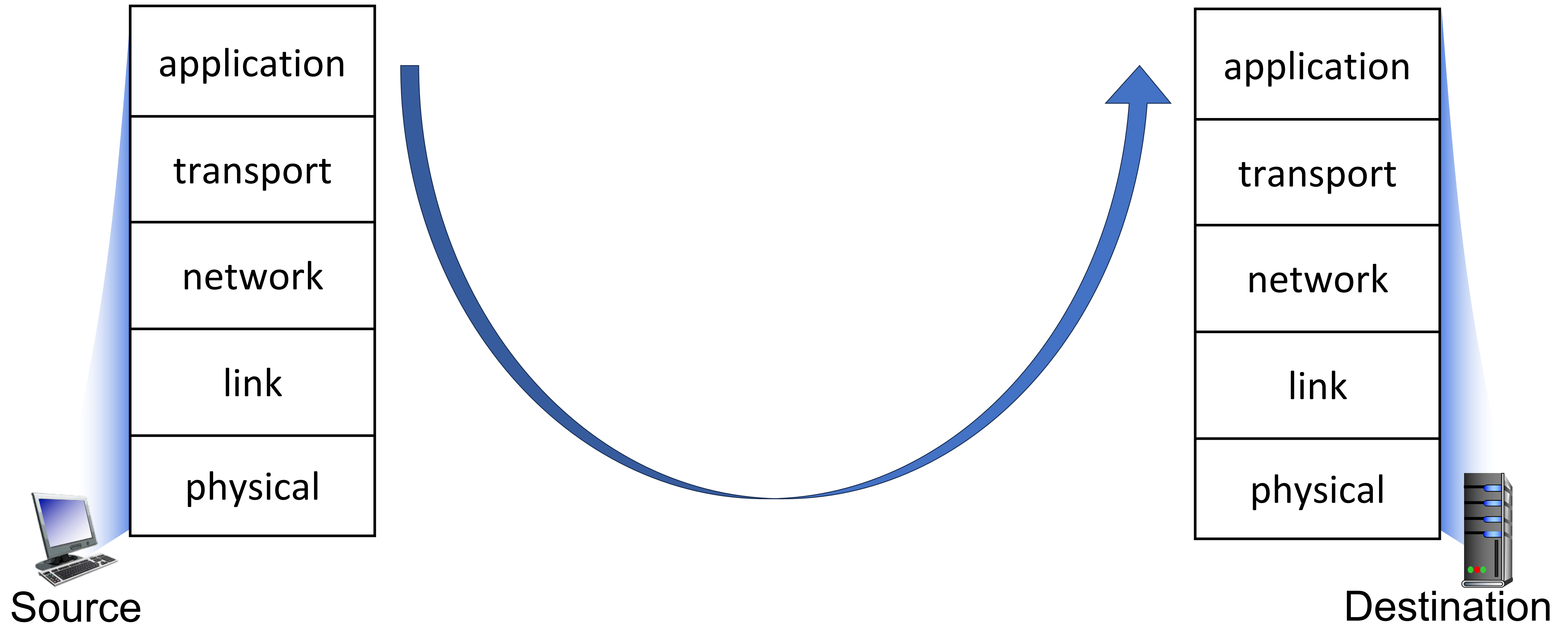
The layered Internet protocol stack

- ▶ **Application Layer:** Supporting Network Applications
 - ▶ HTTP, IMAP, SMTP, DNS
- ▶ **Transport Layer:** Process-process data transfer
 - ▶ TCP, UDP
- ▶ **Network Layer:** Routing of datagrams from source to destination
 - ▶ IP, routing protocols
- ▶ **Link layer:** Data transfer between neighboring network elements
 - ▶ Ethernet, 802.11 (Wifi)
- ▶ **Physical Layer:** Bits "on the wire"

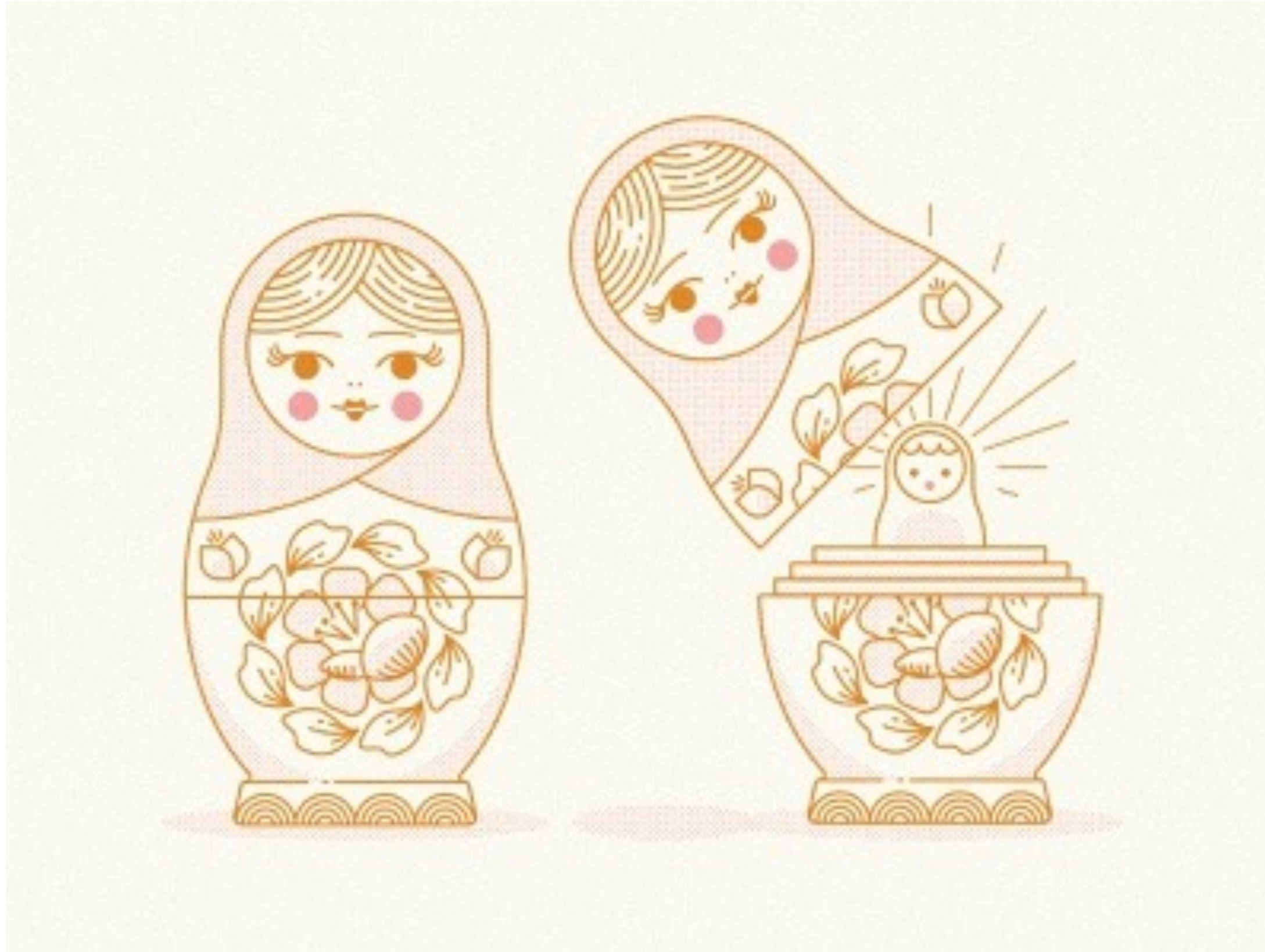


Adapted from slides by J.F Kurose and K.W. Ross

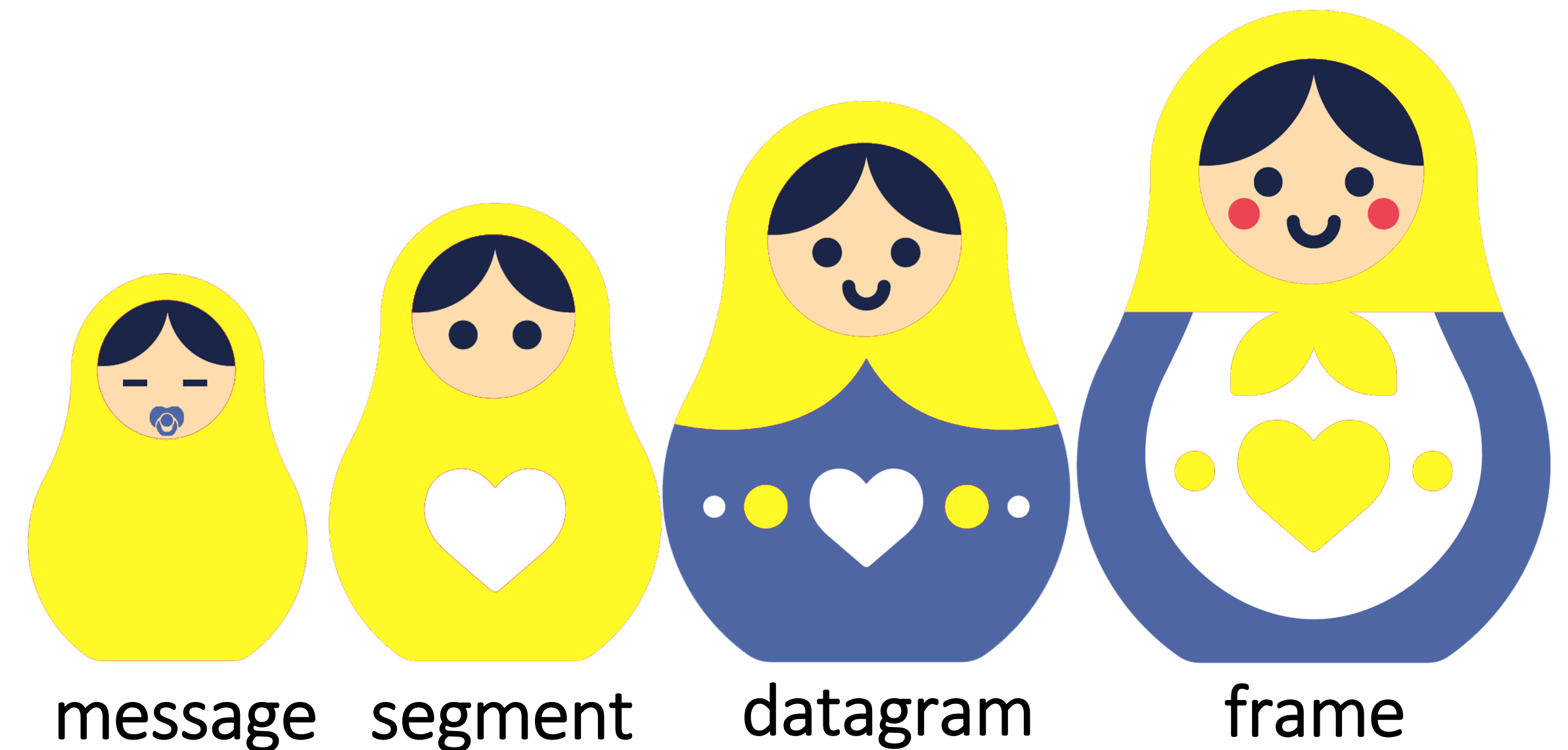
Services, Layering, & Encapsulation



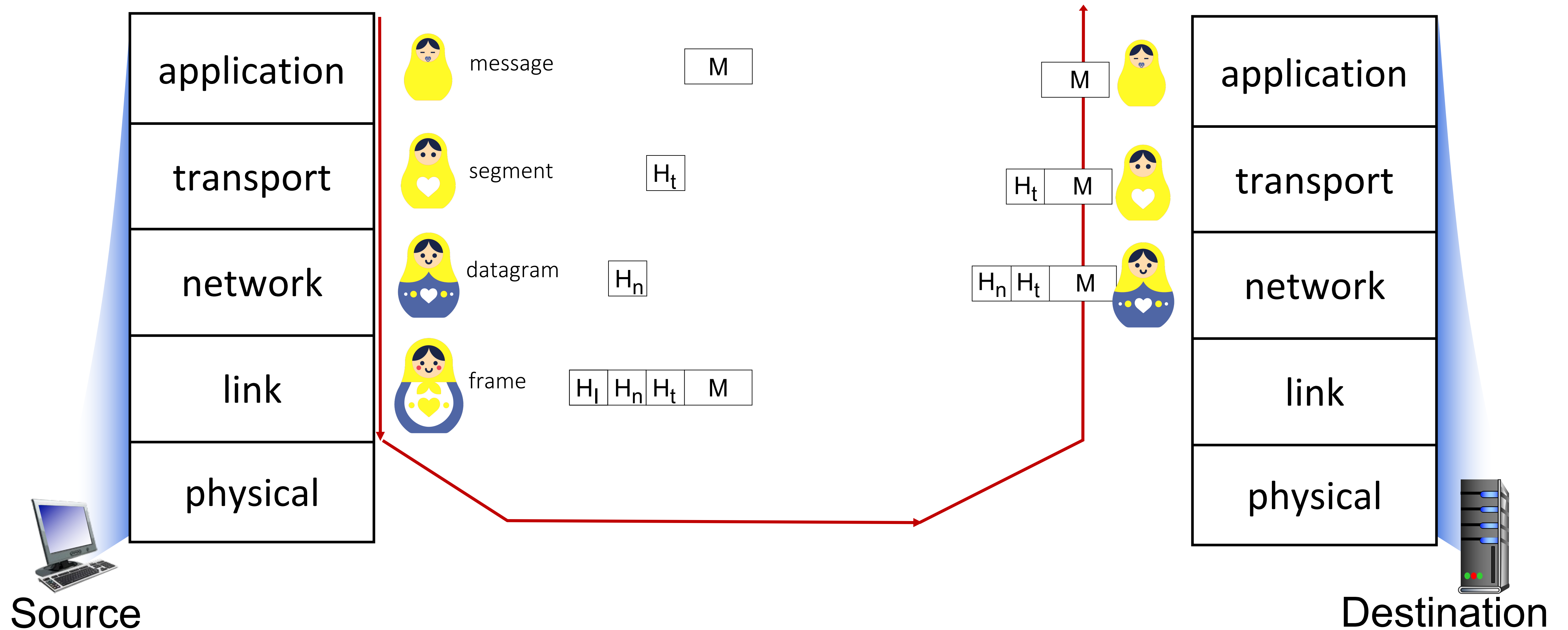
Encapsulation – Think of it like matroyshka dolls



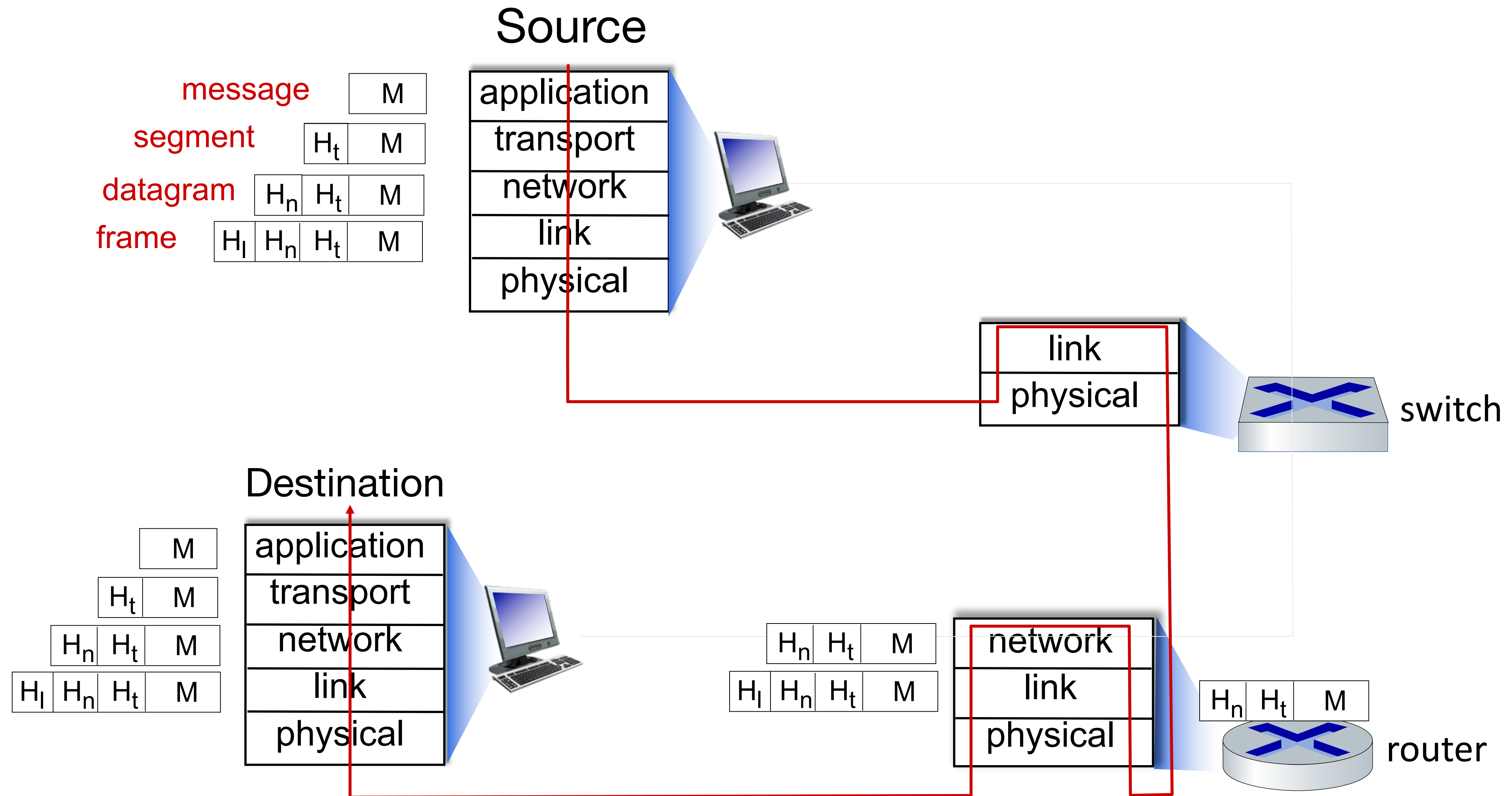
Babushka Boi, <https://dribbble.com/shots/7182188-Babushka-Boi>



Encapsulation – Think of it like matroyshka dolls



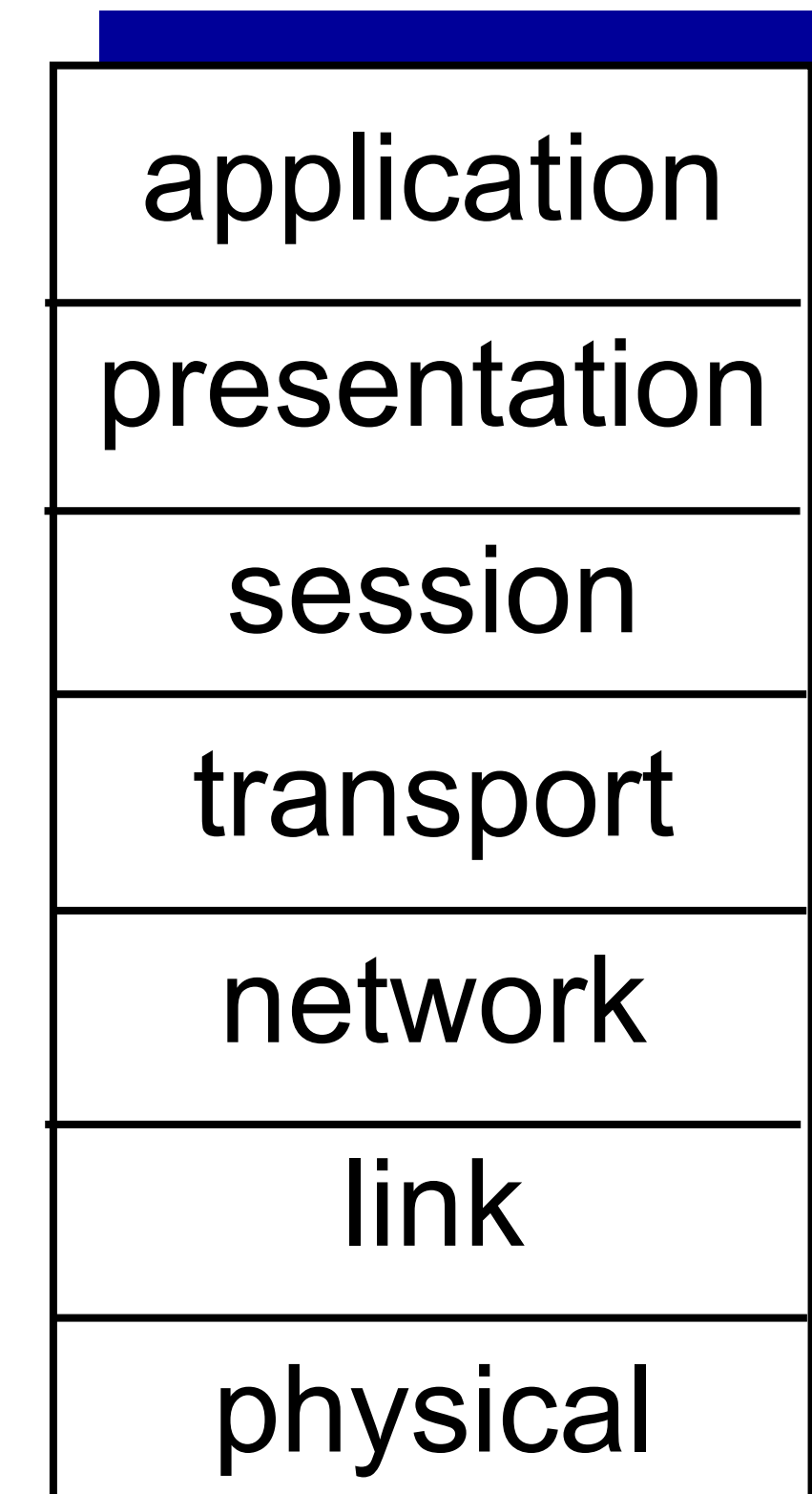
Encapsulation – An end-to-end example



For completeness sake: The ISO/OSI Reference Model

- ▶ The current layered model had a predecessor, the ISO/OSI model
- ▶ This model had two additional layers not found in today's Internet
 - ▶ **Presentation:** allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
 - ▶ **Session:** synchronization, checkpointing, recovery of data exchange
 - ▶ If needed, must be implemented in application

The seven layer OSI/ISO reference model



This will not be in the exam

Adapted from slides by J.F Kurose and K.W. Ross

Learning Goal D7C380

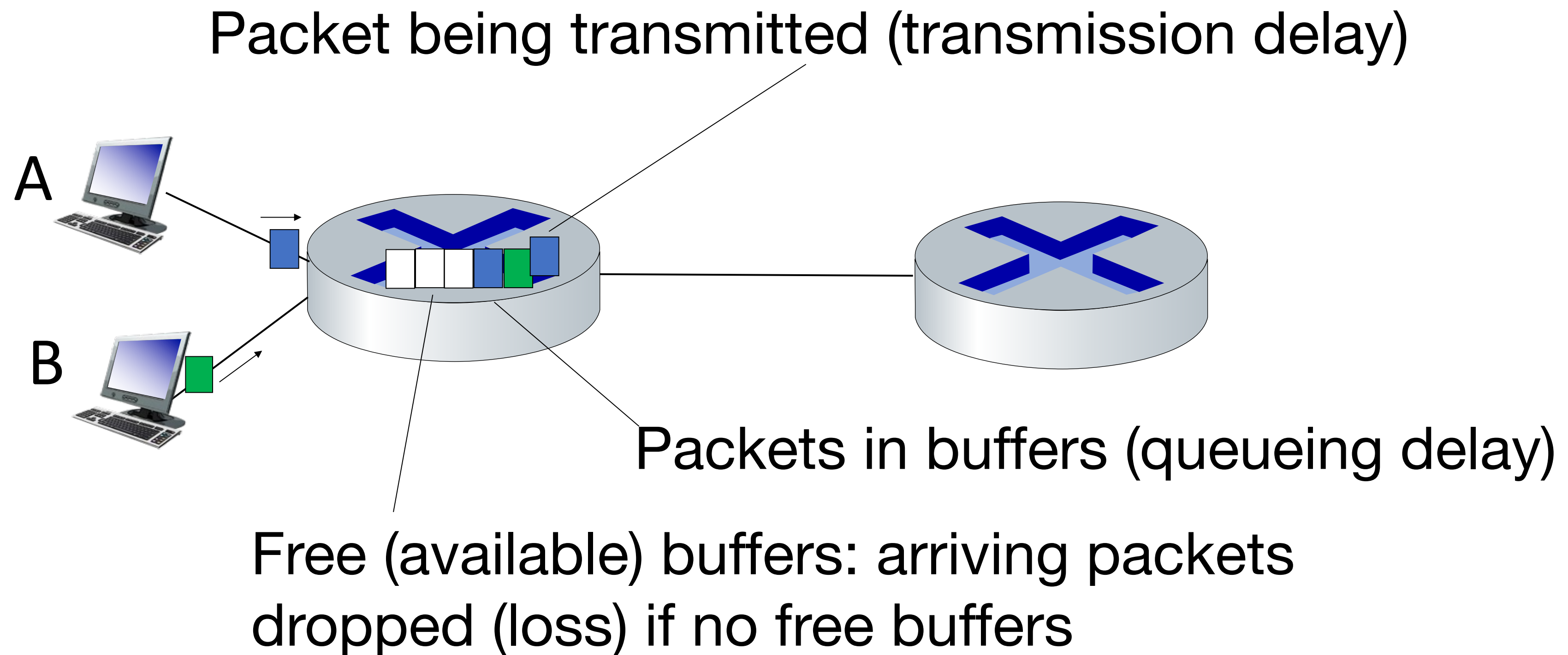
How to measure performance in networks?

Measuring Performance in Computer Networks

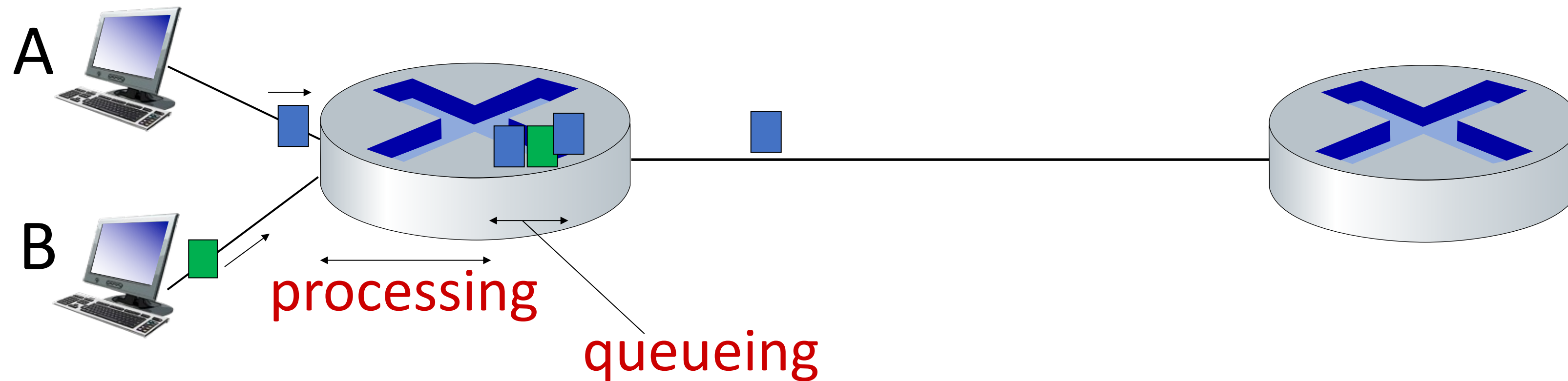
- ▶ Performance of Computer Networks depends on three metrics
- ▶ **Delay:** How long does it take until packets reach the destination
- ▶ **Loss:** What portion of packets are lost along the way
- ▶ **Throughput:** How “space” is available for packets on the connection

How do Packet Delay and Packet Loss occur?

- ▶ Packets queue in router buffers, waiting for turn for transmission
 - ▶ Queue length grows whenever arrival rate to link exceeds output link capacity
- ▶ Packet loss occurs when memory to hold queued packets fills up



The four sources of Packet Delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}}$$

d_{proc} : Processing Delay

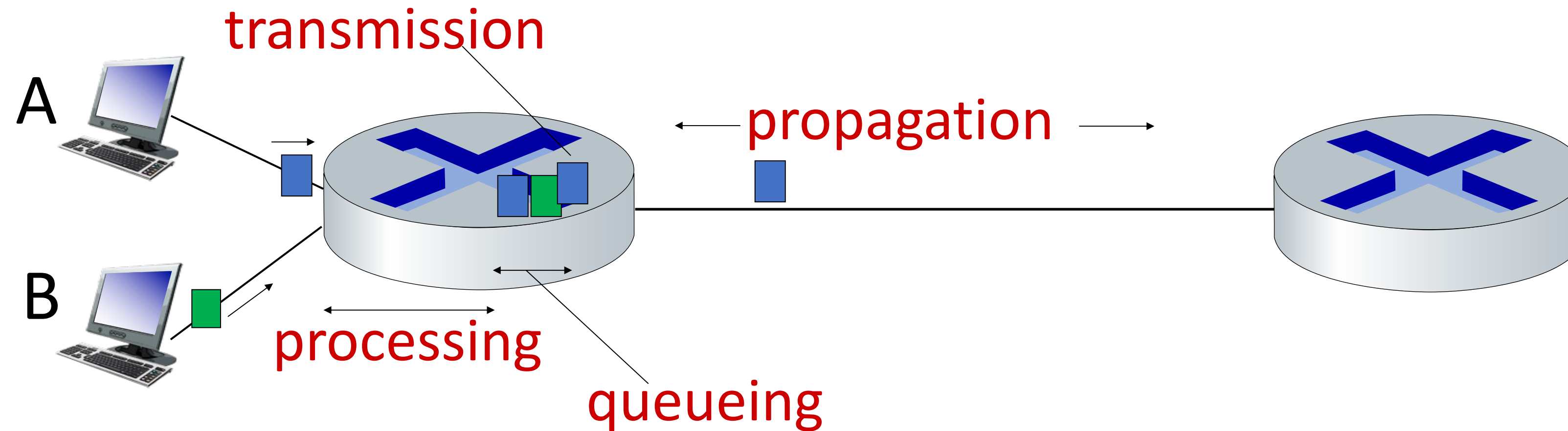
- ▶ Check for bit errors
- ▶ Determine output link
- ▶ Typically < microsecs

Typically negligible

d_{queue} : Queueing Delay

- ▶ Time waiting at output link for transmission
- ▶ Depends on congestion

The four sources of Packet Delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : Transmission Delay

- ▶ L : packet length (bits)
- ▶ R : link transmission rate (bps)

$$d_{\text{trans}} = \frac{L}{R}$$

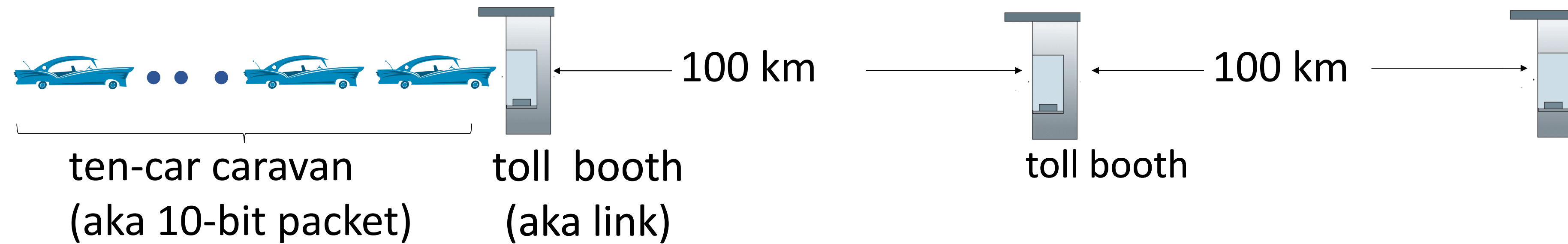
d_{prop} : Propagation Delay

- ▶ d : length of physical link
- ▶ s : propagation speed ($\sim 2 * 10^8 \text{m/sec}$)

$$d_{\text{prop}} = \frac{d}{s}$$

very
different

Transmission vs Propagation Delay: A car analogy



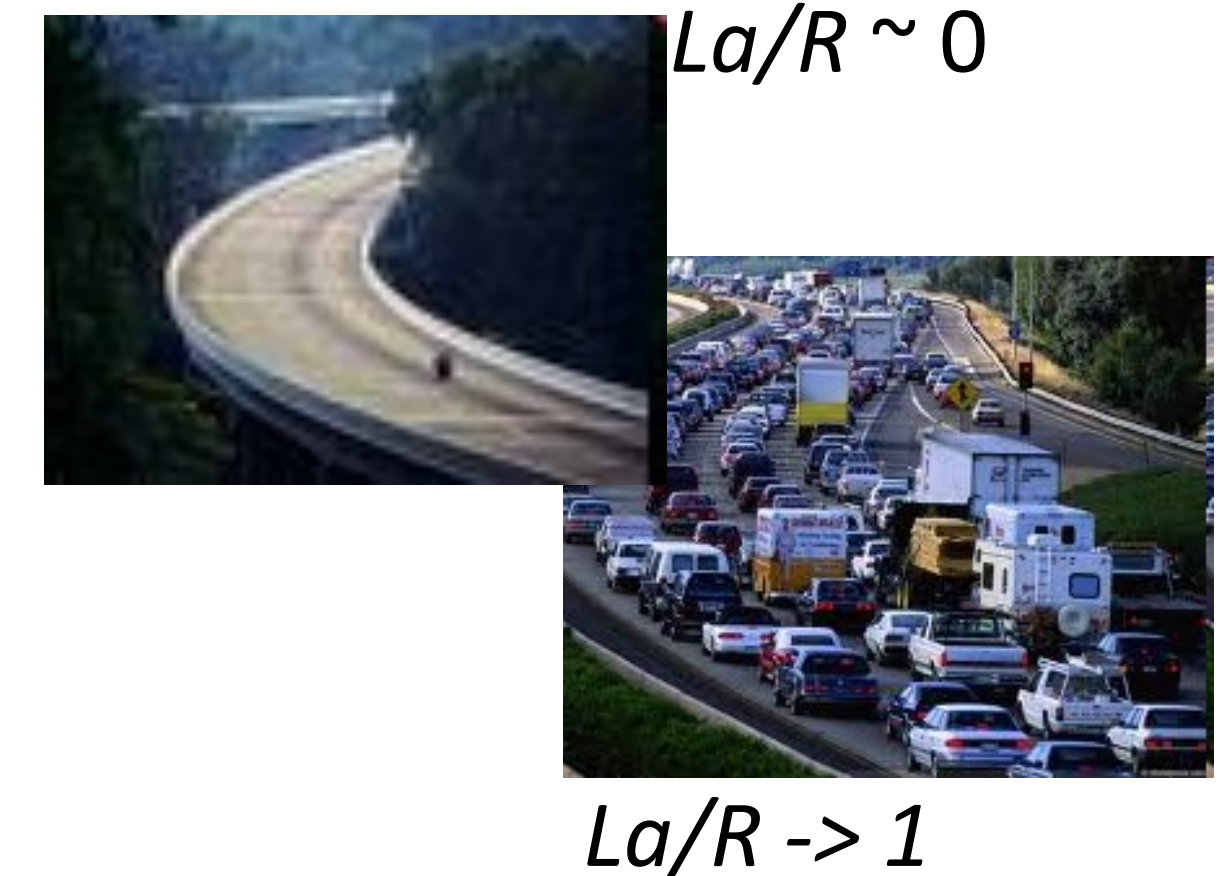
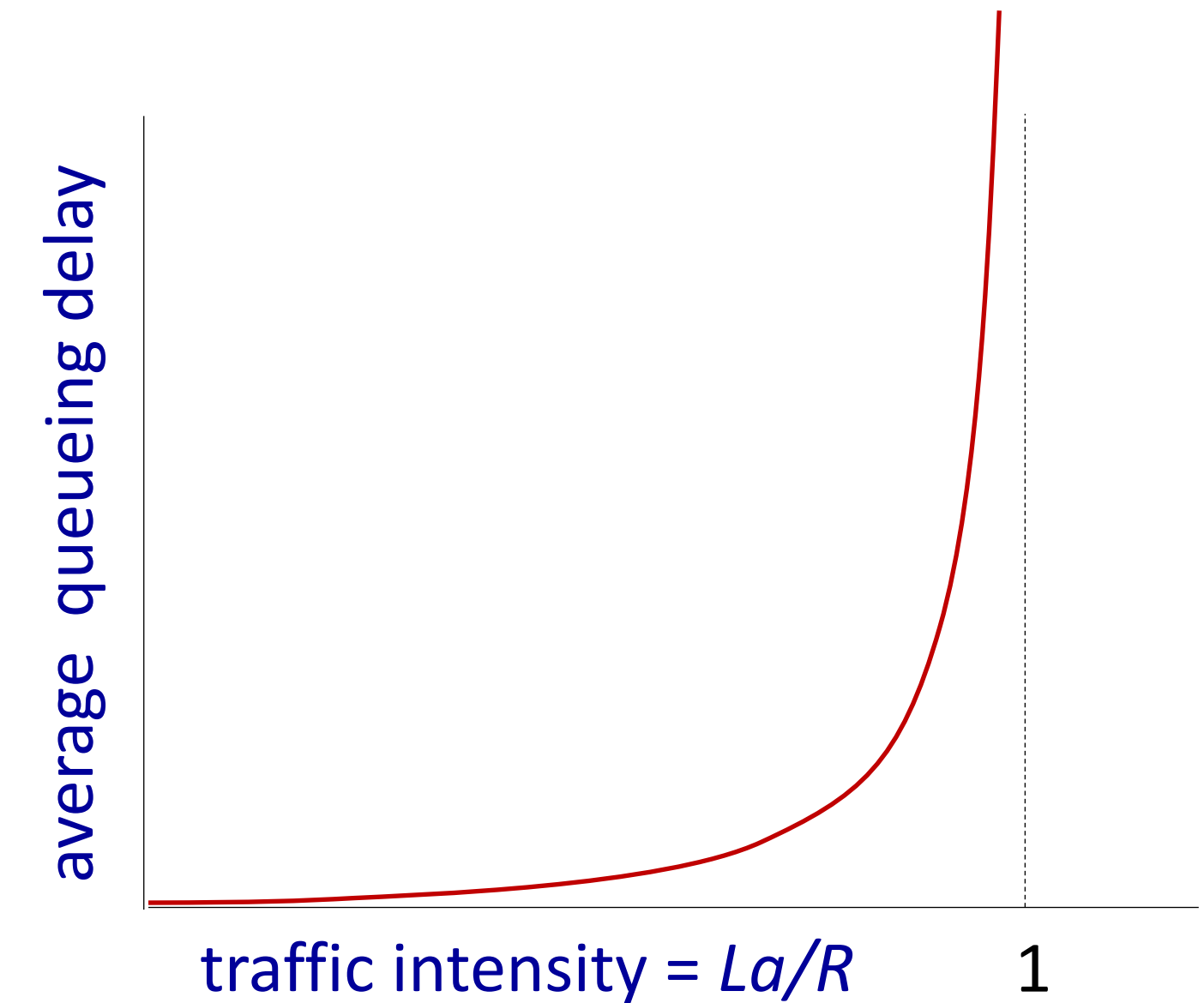
- ▶ Car ~ Bit; Caravan ~ Packet;
Toll service ~ Link transmission
- ▶ Toll booth takes 12 sec to service car (bit transmission time)
- ▶ “Propagate” at 100 km/hr
- ▶ Q: How long until caravan is lined up before 2nd toll booth?
- ▶ Time to “push” entire caravan through toll booth onto highway
 $= 12 \times 10 = 120 \text{ sec}$
- ▶ Time for last car to propagate from 1st to 2nd toll booth:
 $100\text{km} / (100\text{km/hr}) = 1 \text{ hr}$
- ▶ A: 62 minutes

Packet Queueing Delay

- ▶ a : Average packet arrival rate
- ▶ L : Packet length (bits)
- ▶ R : Link bandwidth (bit transmission rate)

$$\frac{L*a}{R} = \frac{\text{arrival rate of bits}}{\text{service rate of bits}} \quad \text{“traffic intensity”}$$

- ▶ $La/R \sim 0$: avg. queueing delay small
- ▶ $La/R \rightarrow 1$: avg. queueing delay large
- ▶ $La/R > 1$: more “work” arriving is more than can be serviced - average delay infinite!



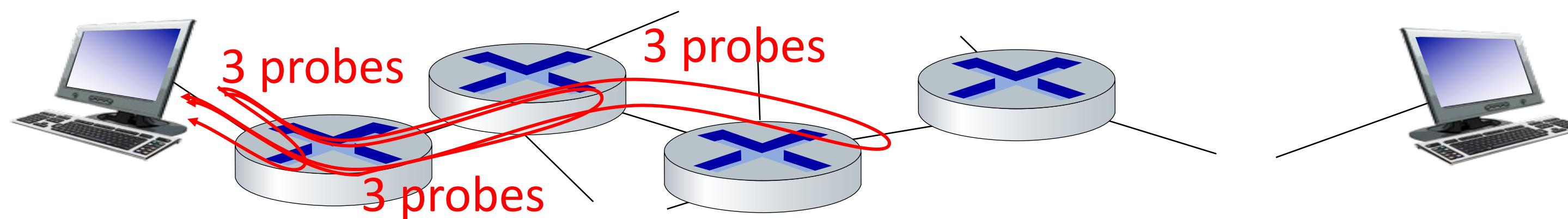
Adapted from slides by J.F. Kurose and K.W. Ross

“Real” Internet delays and routes

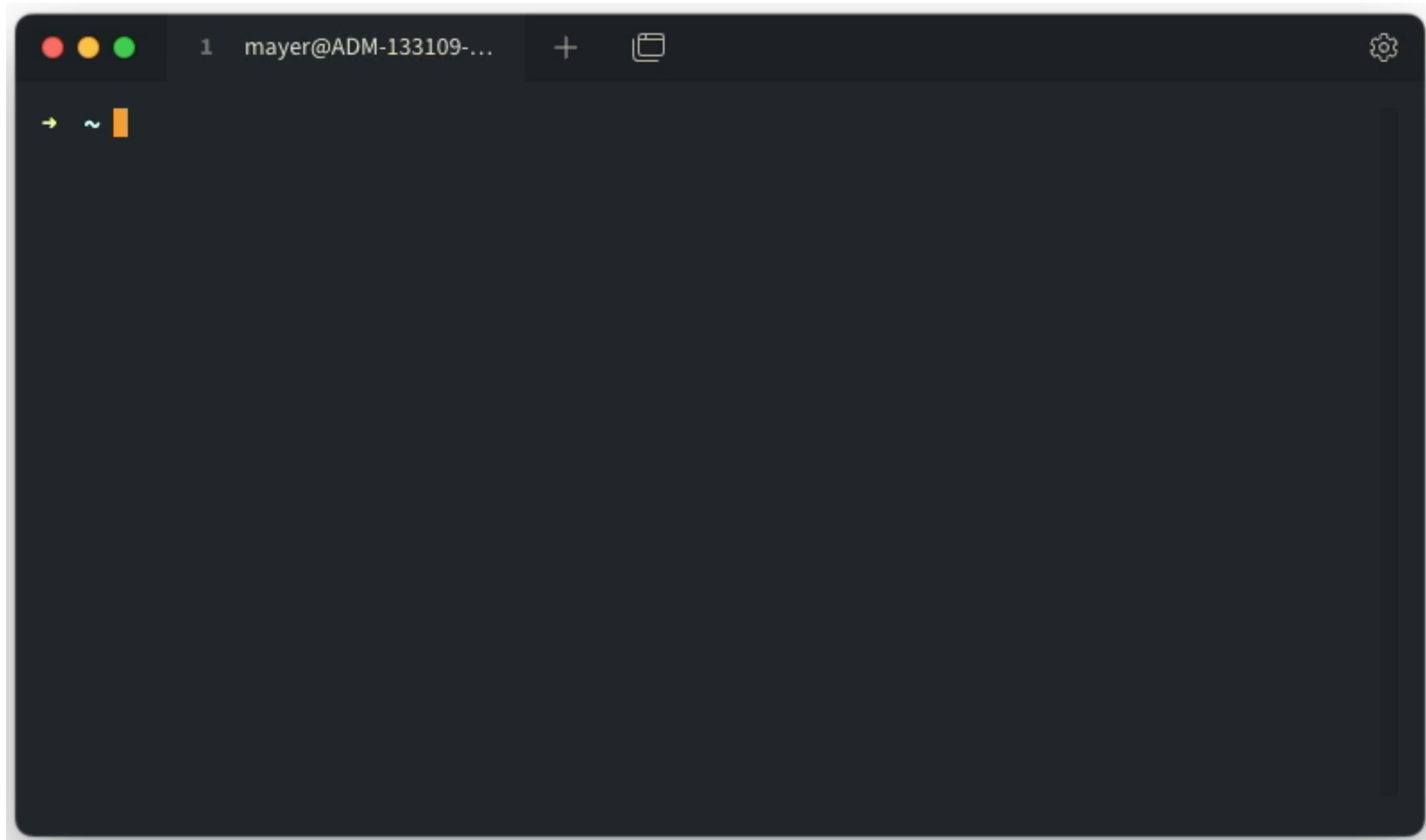
- ▶ What do “real” Internet delay & loss look like?
- ▶ Traceroute program: provides delay measurement from source to router along end-end Internet path towards destination.

For all i :

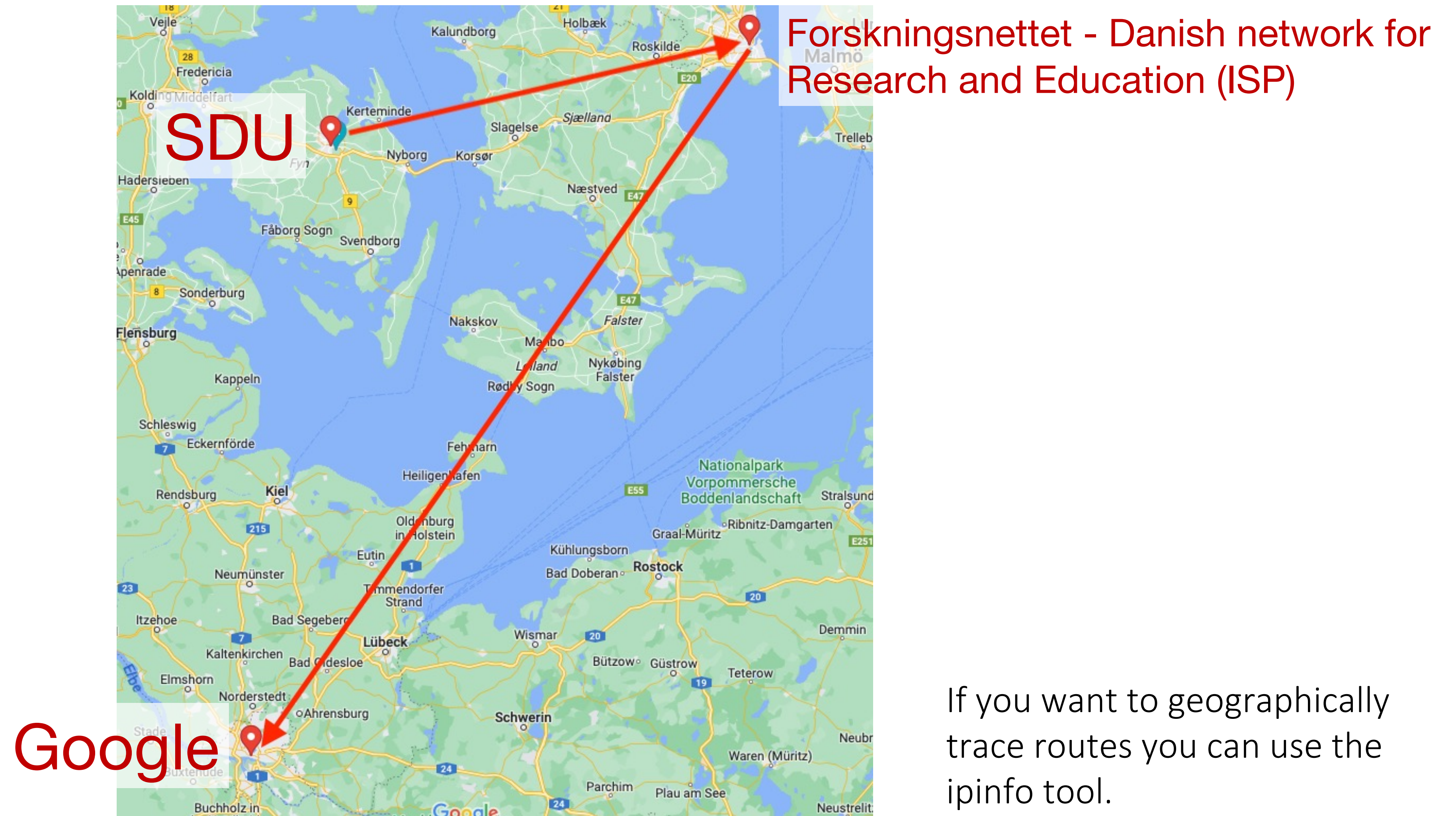
- ▶ sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
- ▶ router i will return packets to sender
- ▶ sender measures time interval between transmission and reply



“Real” Internet delays and routes



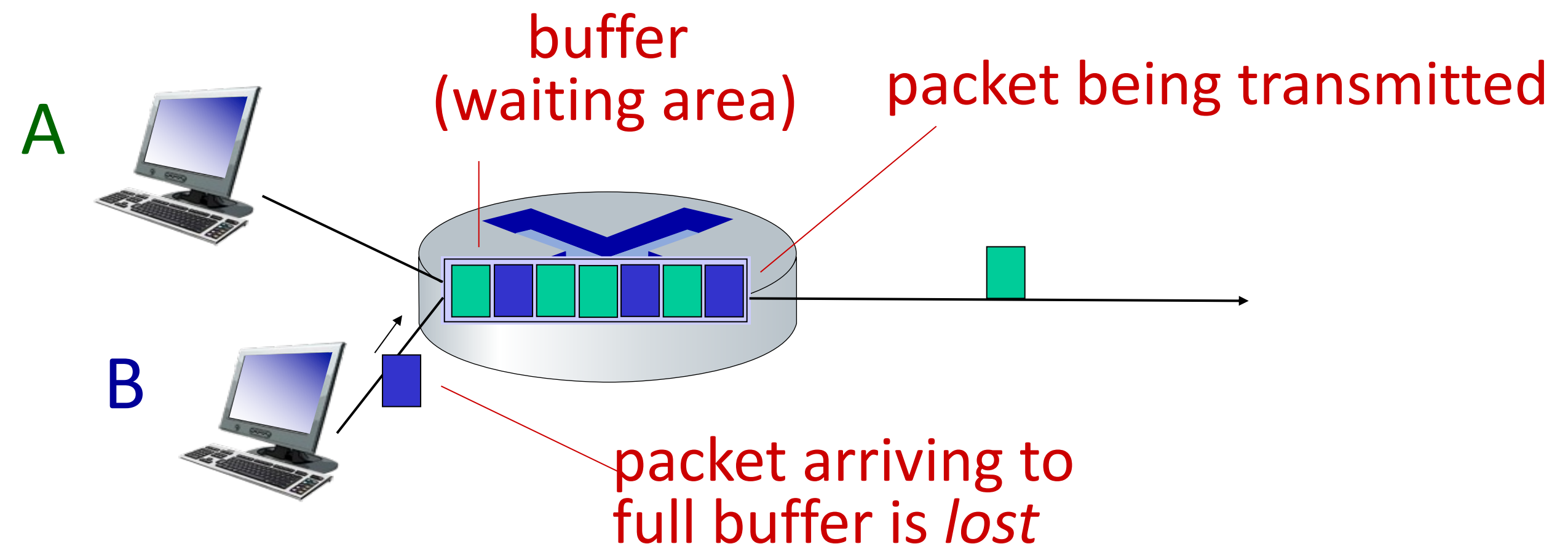
“Real” Internet delays and routes



If you want to geographically trace routes you can use the ipinfo tool.

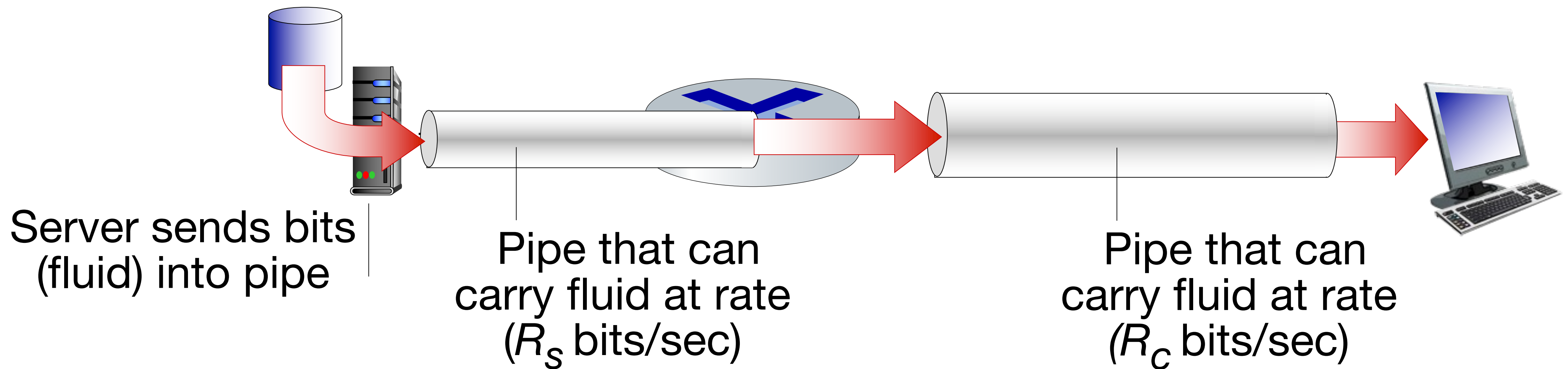
Packet Loss

- ▶ Queue (aka buffer) preceding link in buffer has finite capacity
- ▶ Packet arriving to full queue dropped (aka lost)
- ▶ Lost packet may be retransmitted by
 - ▶ Previous node...
 - ▶ ...by source end system...
 - ▶ ...or not at all



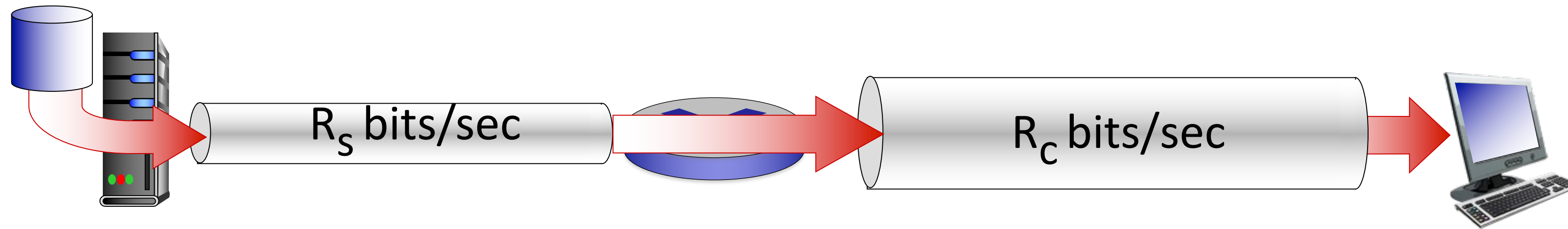
Throughput

- ▶ Throughput: rate (bits/time unit) at which bits are being sent from sender to receiver
- ▶ Instantaneous: rate at given point in time
- ▶ Average: rate over longer period of time

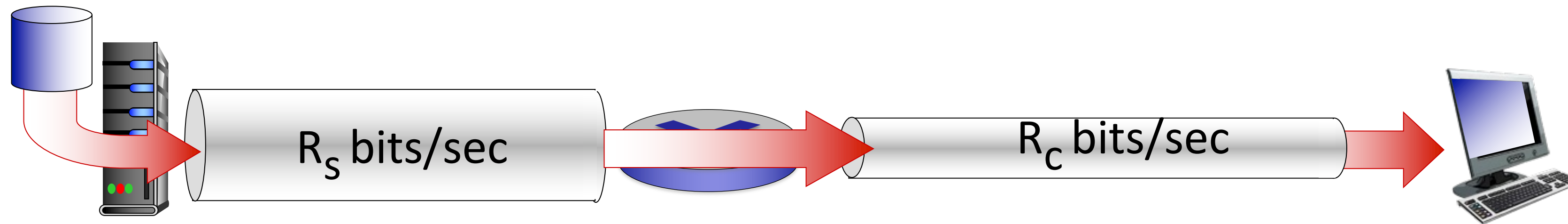


Throughput

$R_s < R_c$: What is average end-to-end throughput?



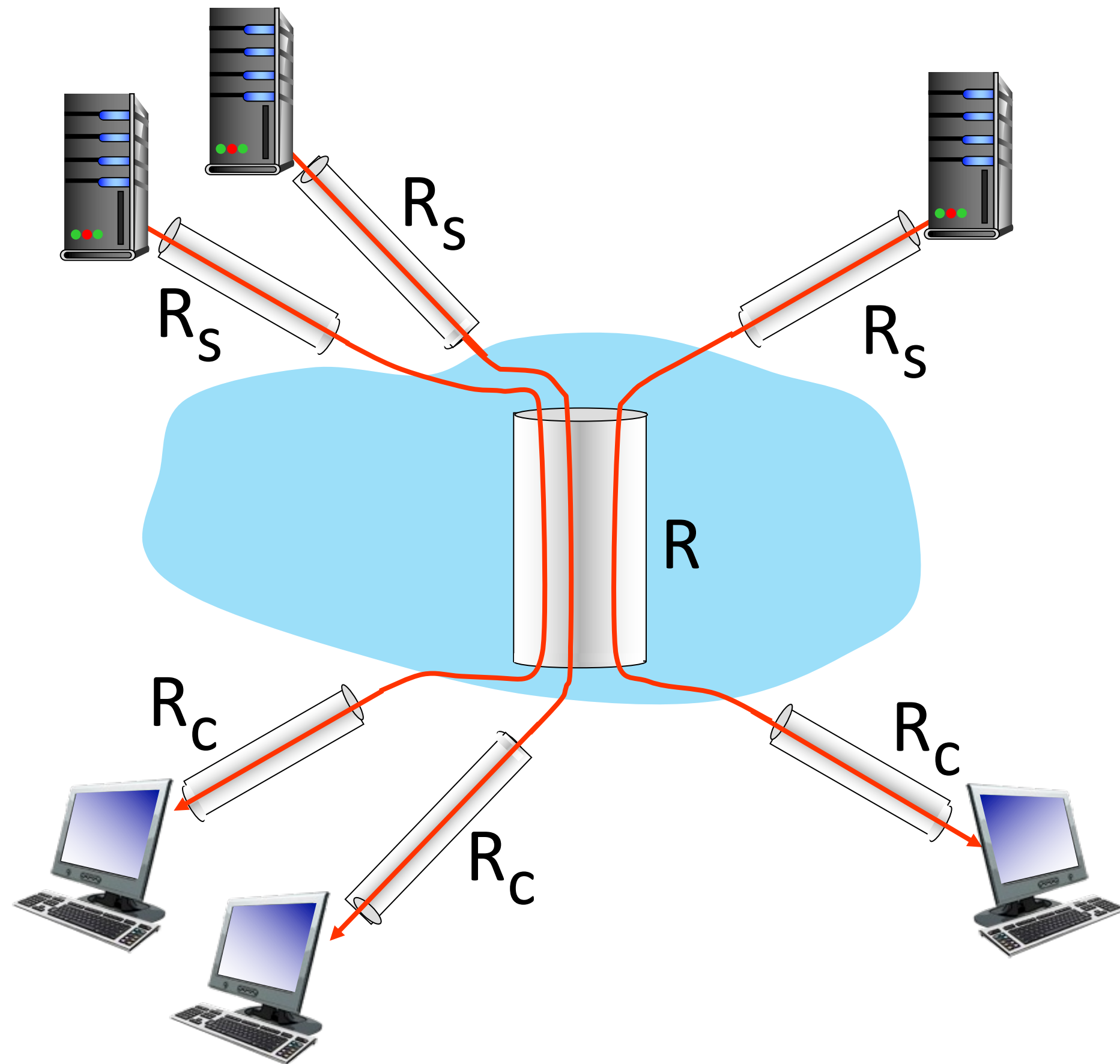
$R_s > R_c$: What is average end-to-end throughput?



Bottleneck link

Link on end-end path that constrains end-end throughput.

Throughput – A network scenario



10 connections (fairly) share
backbone bottleneck link R bits/sec

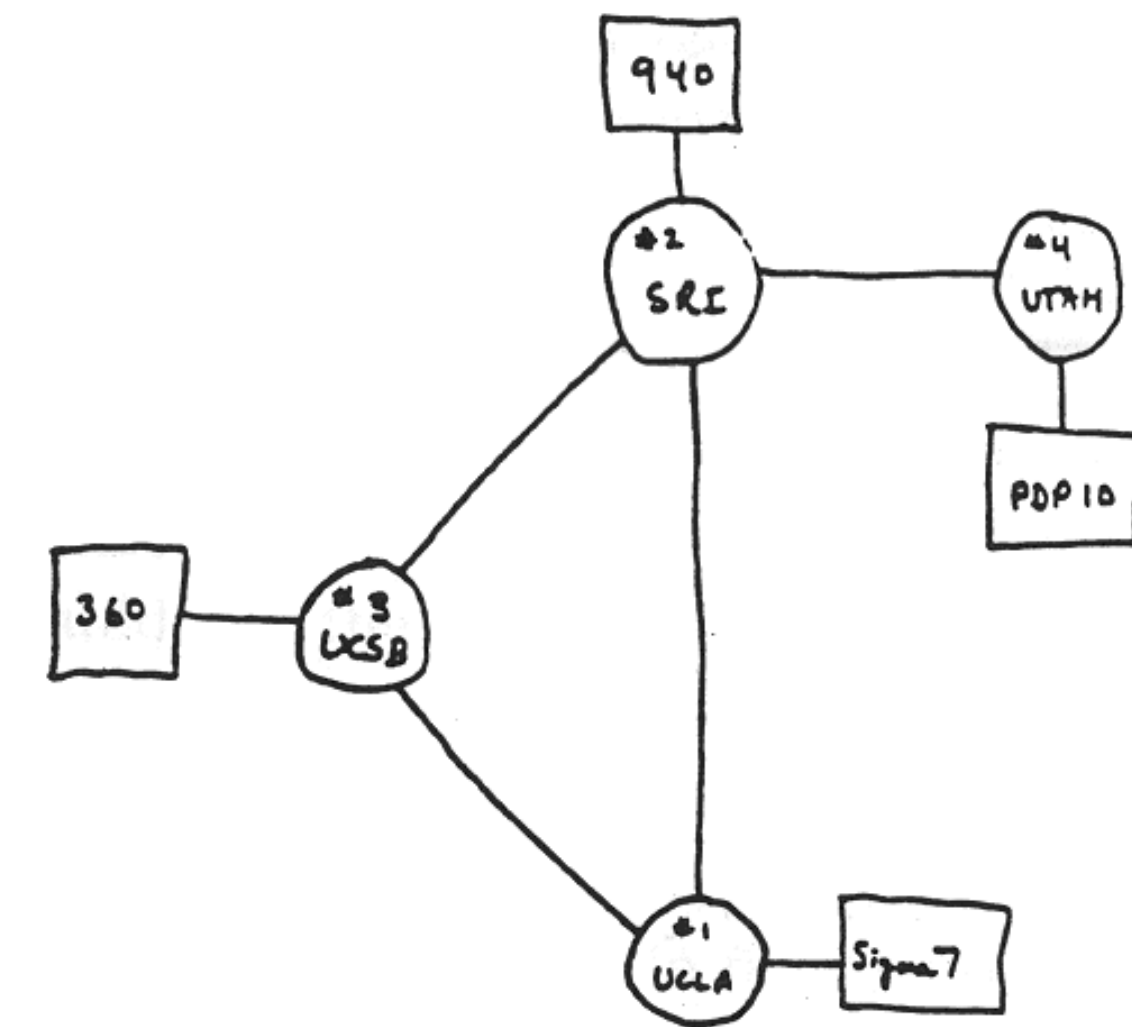
- ▶ Per-connection end-to-end throughput: $\min(R_c, R_s, \frac{R}{10})$
- ▶ In practice: R_c or R_s is often bottleneck

Learning Goal F32ECC

How did the Internet evolve?

A brief history of the Internet – 1961-1972

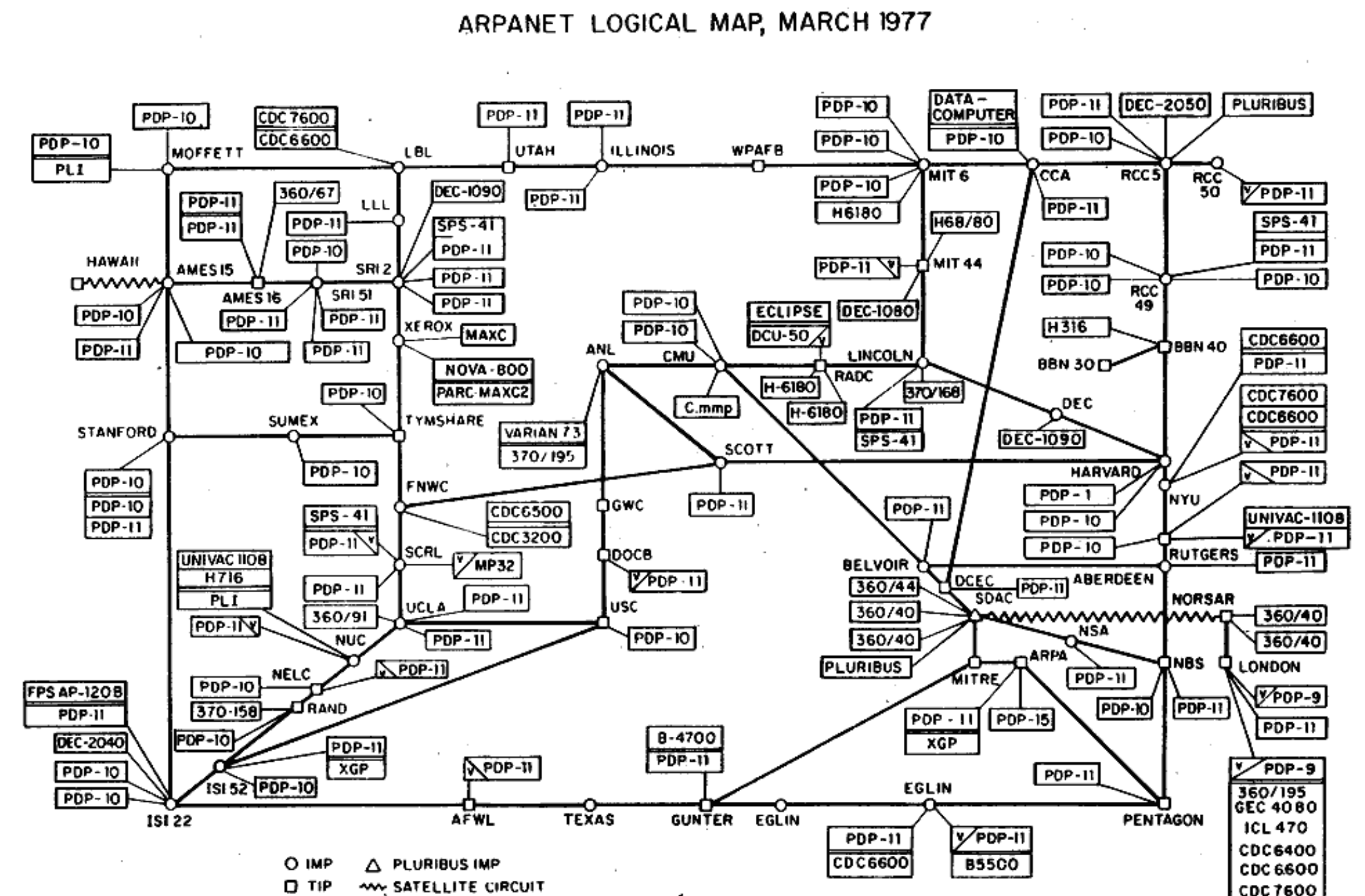
- ▶ 1967: ARPAnet conceived by Advanced Research Projects Agency
- ▶ 1969: first ARPAnet nodes operational
- ▶ 1972:
 - ▶ ARPAnet public demo
 - ▶ NCP (Network Control Protocol) first host-host protocol
 - ▶ First e-mail program



THE ARPA NETWORK

A brief history of the Internet – 1961-1972

- ▶ 1967: ARPAnet conceived by Advanced Research Projects Agency
- ▶ 1969: first ARPAnet nodes operational
- ▶ 1972:
 - ▶ ARPAnet public demo
 - ▶ NCP (Network Control Protocol) first host-host protocol
 - ▶ First e-mail program
 - ▶ ARPAnet has 37 nodes



(PLEASE NOTE THAT WHILE THIS MAP SHOWS THE HOST POPULATION OF THE NETWORK ACCORDING TO THE BEST INFORMATION OBTAINABLE, NO CLAIM CAN BE MADE FOR ITS ACCURACY)
NAMES SHOWN ARE IMP NAMES, NOT (NECESSARILY) HOST NAMES

A brief history of the Internet – 1972-1990

- ▶ 1974: Cerf and Kahn - architecture for interconnecting networks
- ▶ 1979: ARPAnet has 200 nodes
- ▶ 1983: Deployment of TCP/IP
- ▶ 1982: SMTP e-mail protocol defined
- ▶ 1983: DNS defined for name-to-IP-address translation
- ▶ 1985: FTP protocol defined
- ▶ 1988: TCP congestion control

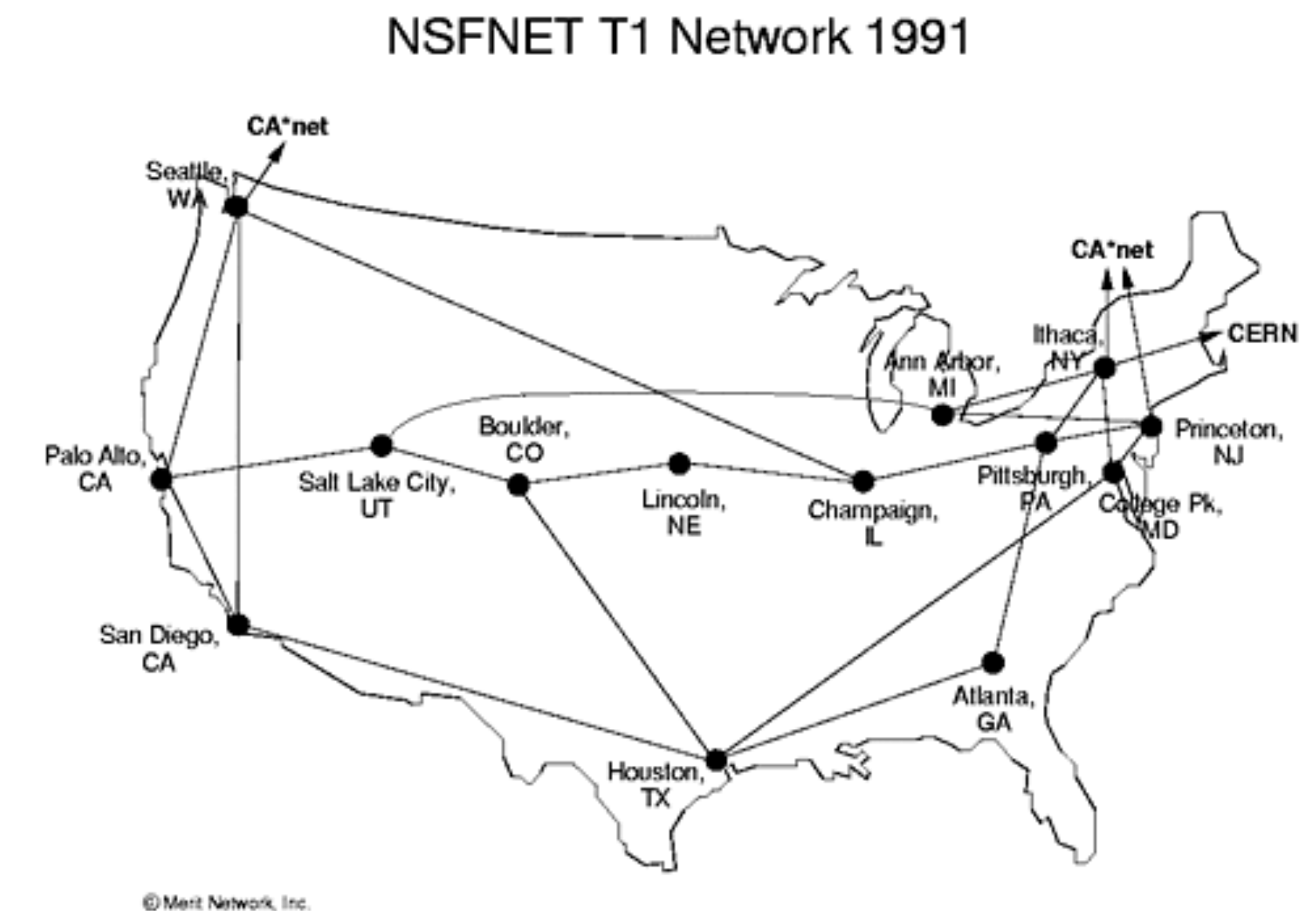
Cerf and Kahn's internetworking principles

- Minimalism, Autonomy - no internal changes required to interconnect networks
- Best-effort service model
- Stateless routing
- Decentralized control

Define today's Internet architecture!

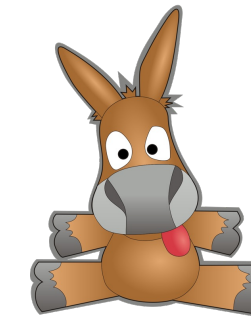
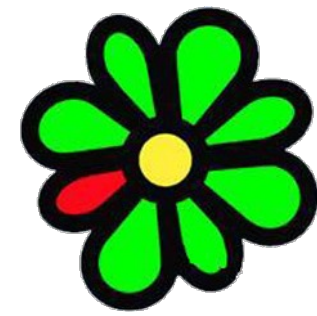
A brief history of the Internet – 1972-Early 1990s

- ▶ Early 1990s: ARPAnet gets replaced with other networks, 100,000 hosts connected to confederation of networks
- ▶ Early 1990s: Web
 - ▶ Hypertext [Bush 1945, Nelson 1960's]
 - ▶ HTML, HTTP: Berners-Lee
 - ▶ 1994: Mosaic, later Netscape
 - ▶ Late 1990s: commercialization of the Web



A brief history of the Internet – Late 1990s-2005

- ▶ More apps: instant messaging, P2P file sharing



- ▶ Platforms that we know until today emerge: Google, ebay, Yahoo, etc.

YAHOO!

Google!

ebay

- ▶ Network security to forefront
- ▶ Est. 50 million hosts, 100 million+ users
- ▶ Backbone links running at Gbps

A brief history of the Internet – 2005-present

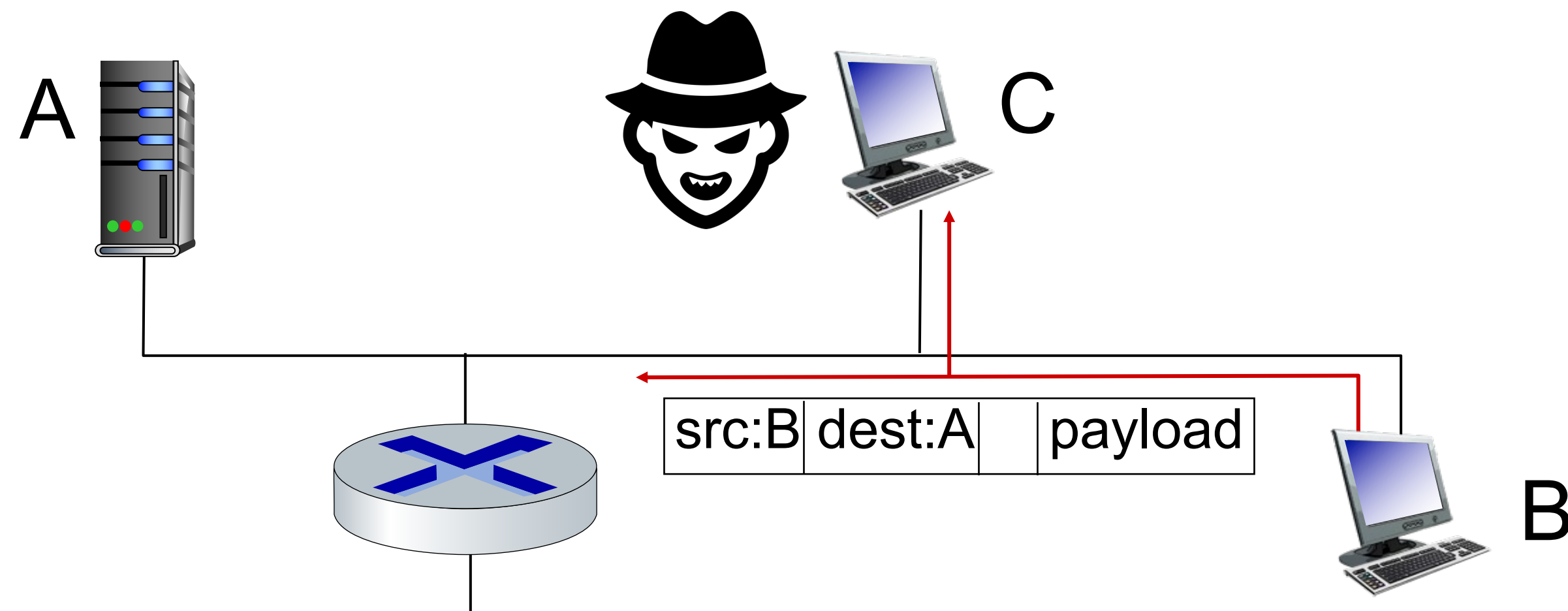
- ▶ Aggressive deployment of broadband home access (10-1000's Mbps)
- ▶ 2008: software-defined networking (SDN)
- ▶ Increasing ubiquity of high-speed wireless access: 4G/5G, WiFi
- ▶ Service providers (Google, FB, Microsoft) create their own networks
- ▶ Bypass commercial Internet to connect “close” to end user, providing “instantaneous” access to social media, search, video content, ...
- ▶ Enterprises run their services in “cloud”
- ▶ Rise of smartphones: more mobile than fixed devices on Internet
- ▶ ~19.8B devices attached to Internet (2025, statista.com)

Implications of the early designs

- ▶ Internet not originally designed with (much) security in mind
 - ▶ Original vision: “a group of mutually trusting users attached to a transparent network” → Not what we have today 🤔
 - ▶ Internet protocol designers playing “catch-up”
 - ▶ Security considerations in all layers!
- ▶ We now need to think about:
 - ▶ How bad guys can attack computer networks
 - ▶ How we can defend networks against attacks
 - ▶ How to design architectures that are resilient to attacks
- ▶ What are typical attacks in computer networks?

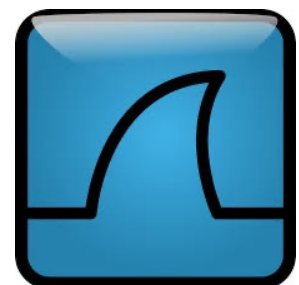
Typical Attack – Packet Sniffing

- ▶ Intercepting packets in broadcast media (shared Ethernet, wireless)
- ▶ Promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



Q: Which STRIDE attack type is the interception of a password?

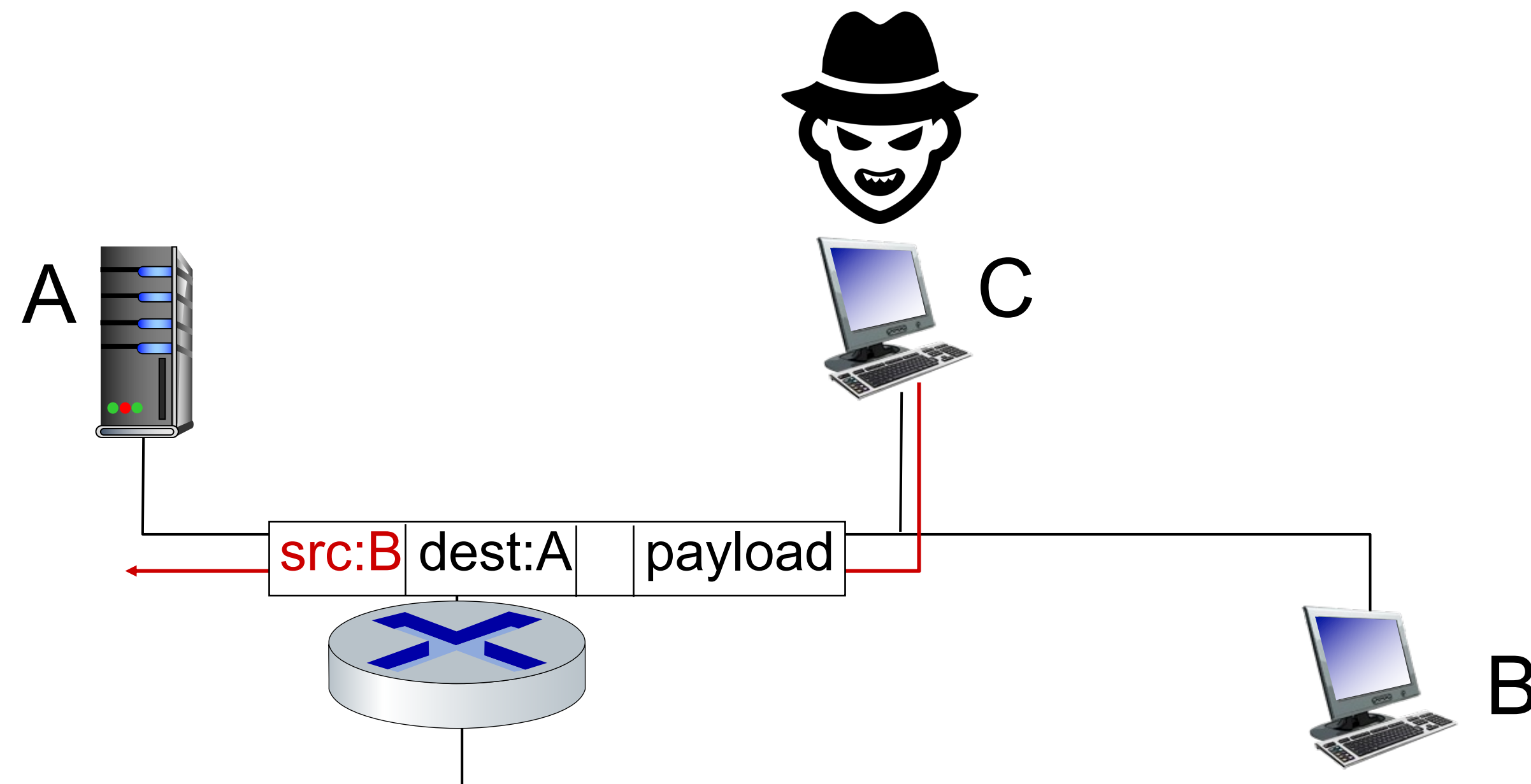
A: Information Disclosure



Wireshark software used for exercises is a (free) packet-sniffer

Typical Attack – IP Spoofing

- ▶ Packet injection with false source address



Q: Which STRIDE attack type is this?

A: Spoofing

Typical Attack – IP Spoofing

- Used mainly to disguise origin of other attacks

NETWORK SECURITY

IP Spoofing Attack Tried to Disrupt Tor Network

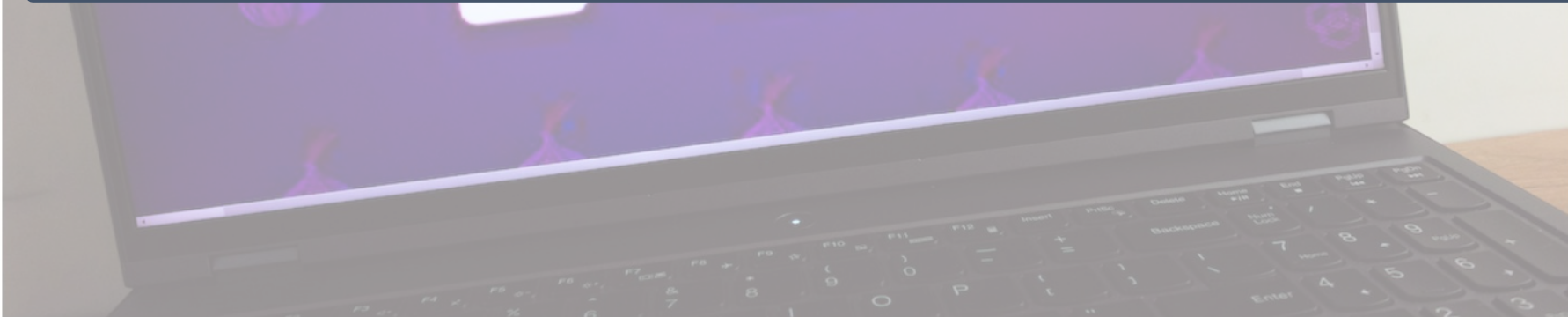
A coordinated IP spoofing attack that involved port scans tried to disrupt the Tor network by getting relays on blocklists.

By  Eduard Kovacs | November 12, 2024 (4:25 AM ET)

The unauthorized port scanning triggered automated abuse complaints to ISPs, which resulted in some relays being taken offline.

An analysis revealed that a threat actor had used spoofed SYN packets to make it appear as if IPs associated with Tor relays had been conducting the port scans. Non-exit relays were the focus of the attack.

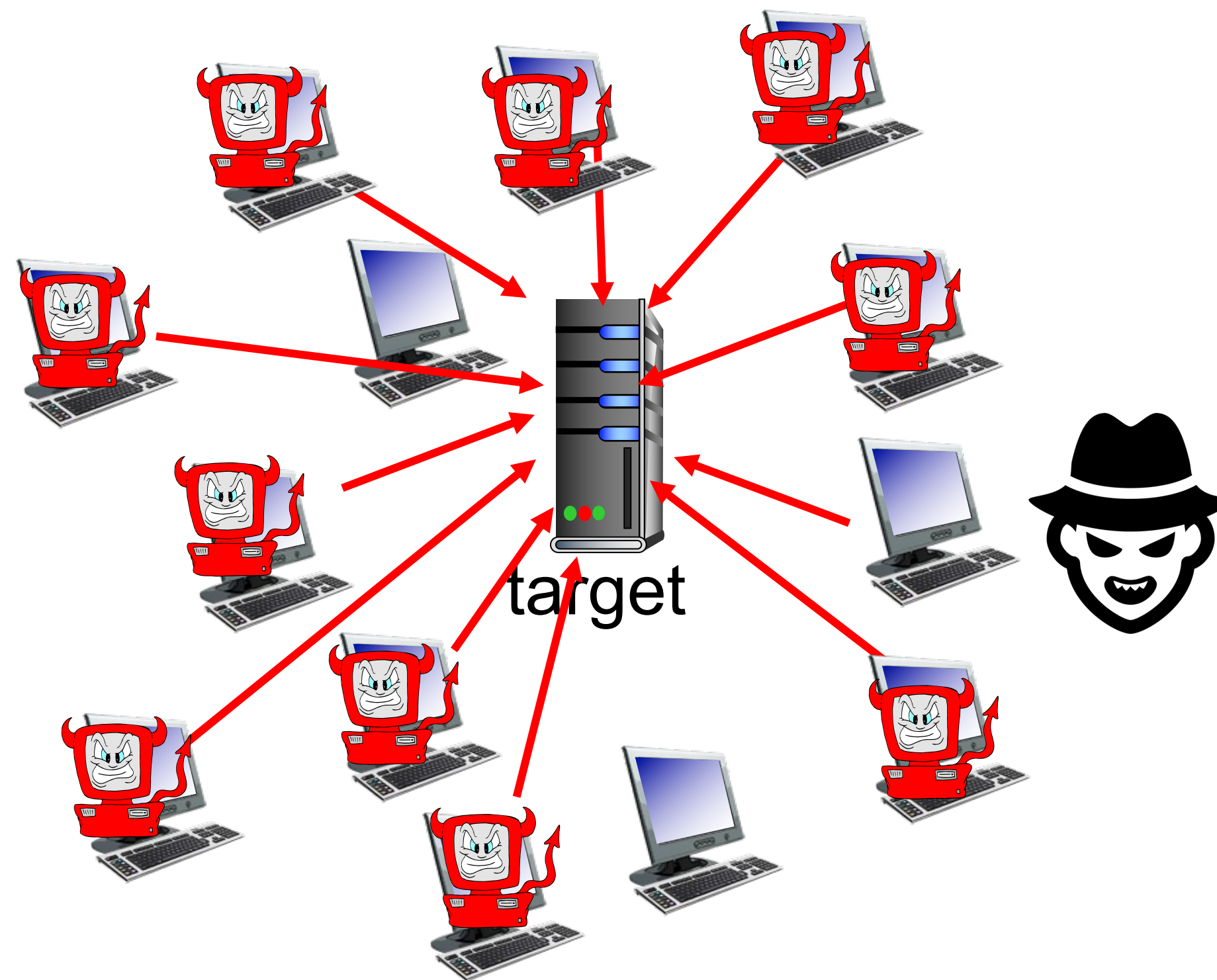


A coordinated IP spoofing attack attempted to disrupt the Tor anonymity network, according to the Tor Project and relay operators.

- Attack
- 5 GitHub Workflows Attack Affects Hundreds of Repos, Thousands of Secrets
- 6 Fintech Firm Wealthsimple Says Supply Chain Attack Resulted in Data Breach
- 7 Salesloft GitHub Account Compromised Months Before Salesforce Attack

Typical Attack – Denial of Service

- ▶ Attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic
- ▶ Potentially using multiple source hosts → Distributed Denial of Service



Q: Which STRIDE attack type is this?

A: Denial of Service

Typical Attack – Denial of Service

► Easy to do so they happen a lot

Danish Banks Are Targets of Pro-Russian DDoS Hacking Group

Nuisance Hacking Group NoName057(16) Takes Credit

Akshaya Asokan ([@asokan_akshaya](#)) • January 10, 2023

 **CLOUDFLARE** Platform Products Developers

Famous DDoS attacks: The largest DDoS attacks of all time

<https://www.cloudflare.com/en-gb/learning/ddos/famous-ddos-attacks/>

Denmark believes Russia is involved Defending the Internet: how Cloudflare blocked a monumental 7.3 Tbps DDoS attack

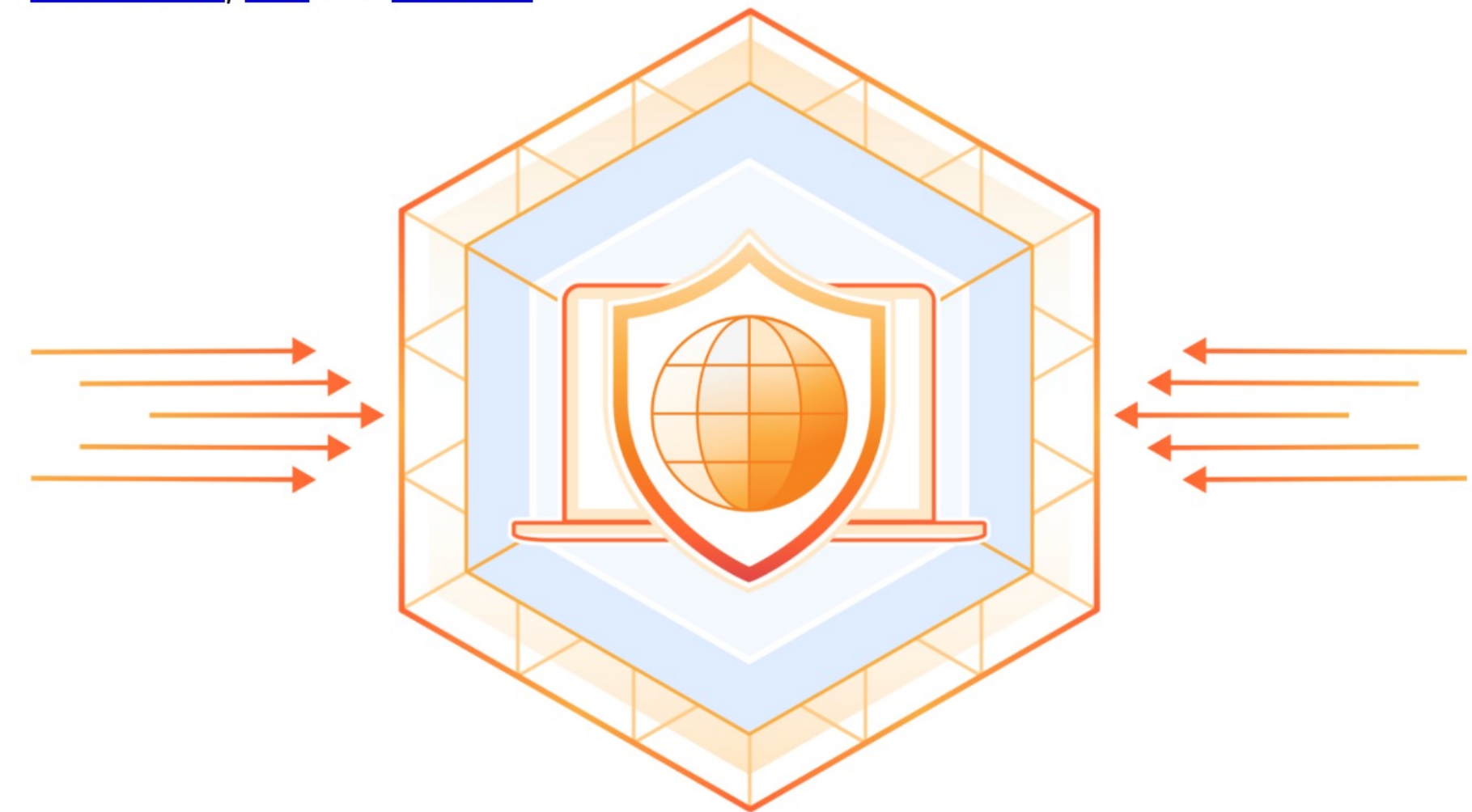
2025-06-19



Omer Yoachimik

9 min read

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websites of Århus, Vejle, Esbjerg, Dragør, Frederiksberg, and other municipalities.

Typical Defenses

- ▶ Authentication, e.g., cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- ▶ Confidentiality, e.g., via encryption
- ▶ Integrity checks, e.g., using digital signatures prevent/detect tampering
- ▶ Access restrictions, e.g., password-protected VPNs
- ▶ Firewalls:
 - ▶ Specialized “middleboxes” in access and core networks
 - ▶ Off-by-default: filter incoming packets to restrict senders, receivers, applications
 - ▶ Detecting/reacting to (D)DOS attacks
- ▶ And many more...

Learning Goals

- ▶ Get an understanding of what a computer network is is ✓
 - ▶ Network Topology
 - ▶ The network edge
 - ▶ Access networks
 - ▶ The network core
 - ▶ Forwarding & Routing
- ▶ Understand what a protocol is ✓
 - ▶ Definition & Example
- ▶ Understand how layers form the structure of the Internet (and other networks) ✓
 - ▶ General concepts of layering
 - ▶ The 5-layer Internet model
 - ▶ Services, Layering, and Encapsulation

Learning Goals

- ▶ Know how to measure the performance of computer networks ✓
 - ▶ Delay
 - ▶ Loss
 - ▶ Throughput
- ▶ Understand the implications of the Internet's history ✓
 - ▶ A brief intro to the history of the Internet
 - ▶ Implications for security